

Engineered Carbon Removals methodology

BeZero Carbon Ratings



BeZero

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Glossary

Baseline: A baseline is the volume of emissions that are assumed to occur in the absence of any project intervention.

BeZero Carbon ex ante Rating: A BeZero Carbon ex ante Rating represents BeZero Carbon's opinion on the likelihood of the carbon credit achieving a tonne of CO₂e avoided or removed. It is an opinion on the greenhouse gas efficacy of an ex ante credit (pre-issuance).

BeZero Carbon Rating (BCR): The BeZero Carbon Rating is a hybrid analytical framework that fuses financial, scientific, and policy-based analysis. It provides an opinion on the likelihood that a given credit achieves a tonne of CO₂e avoided or removed.

Biochar: Projects that produce biochar via pyrolysis, the partial combustion of sustainable biomass, with CO₂ subsequently stored in soil or non-soil applications.

Bioenergy With Carbon Capture & Storage (BECCS): Projects that produce bioenergy from the combustion of sustainable biomass and geologically store the emitted CO₂.

Carbon capture, utilisation and storage (CCUS): Projects that capture CO₂ from a point emission source (e.g. fossil fuel energy production) and store it geologically (CCS), or use the CO₂ in the production of products (e.g. fuels, chemicals, concrete) (CCU/CCUS).

Carbon dioxide removal (CDR): *See Engineered Carbon Removals*

Carbon finance: Financial resources allocated to carbon project activities that reduce or avoid greenhouse gas emissions. It is an additional source of revenue that seeks to make such activities more financially viable.

Closed-loop system: Projects that involve a high degree of technical control or containment.

Commitment period: The duration of time over which a project plans its sequestration activities to be maintained.

Direct Air Capture With Carbon Storage (DACCS): Projects that use technology to remove CO₂ from the atmosphere and store it geologically.

Durability: Durability refers to the robustness of carbon storage. Durability is used interchangeably with permanence to define this storage component. BeZero defines durability in our Durability Assessment.

Durability Assessment: The [2023 BeZero Carbon Durability Assessment](#) evaluates the relative durability of seven carbon removal storage methods, assessing their physical storage, risk of reversal, and scientific certainty.

Engineered Carbon Removals: Engineered technical solutions that enhance or facilitate the removal of greenhouse gases from the atmosphere. One of six sector groups in the BeZero Sector Classification.

Enhanced Weathering: Projects that accelerate the natural process of rock weathering, which stores atmospheric CO₂ in carbonate minerals

Ex ante credits: Carbon credits that are not yet issued, i.e. pre-issuance.

Ex post credits: Carbon credits that have been issued, i.e. post-issuance.

Greenfield site: Land that is being considered for development that has not previously been built on.

Greenhouse gas removals (GGR): *See Engineered Carbon Removals*

Methodology: A framework that sets out a process and parameters for quantifying the carbon benefits of a project. Methodologies enable project developers to quantify baselines, assess key aspects like additionality, and define necessary monitoring practices to ensure the greenhouse gas emission reductions are verifiable, additional, and quantified.

Nature-Based Solutions (NBS): Nature-Based Solutions projects aim to achieve emission removals or avoidance in the natural environment, these include forestry, soil carbon, and blue carbon.

Open-loop system: Projects that involve processes outside technical control or containment.

Project area: The geographical boundaries within which project activities take place.

Pyrolysis: A combustion process that occurs at high temperatures and in the absence of oxygen, which thermochemically converts organic materials into a solid residue with higher carbon content and liquid or gas products mainly containing hydrocarbon components. Used in biochar and bio-oil methods.

Rating Committee: A group of members with extensive experience within and across all sectors who discuss and debate the assignment of ratings. The Rating Committee oversees the analytical process and ensures consistency in the application of rating criteria, methodologies, and comparability of ratings across sectors and countries.

Scalability Assessment: The [2023 BeZero Carbon Scalability Assessment](#) is a first-of-its-kind assessment that evaluates the relative scalability of five Engineered Carbon Removals. The framework compares these methods against eight barriers to scalability (Energy, Land, Finance, Policy, Resources, Localised Impacts, Ancillary Value Chains, and Monitoring, Reporting & Verification).

Stakeholders: Individuals, and/or groups of people who may be affected by the implementation and activities of a project.

Standards bodies: Organisations that develop and revise methodologies to determine the greenhouse gas benefits of various types of carbon projects, and inform when and how many credits projects can issue.

Vintage: The vintage year refers to the period of time when the emission reduction or removal associated with a credit occurred.

Voluntary carbon market (VCM): Carbon markets that exist outside of the regulation and frameworks of international agreements, and are fuelled by commitments of individuals, as well as private and public organisations to reduce GHG emissions through the trading of carbon credits.

Introduction

BeZero Carbon is a global ratings agency for the voluntary carbon market. Our ratings are a publicly available, risk-based framework for assessing carbon efficacy. We rate qualifying carbon credits in all sectors.

We take a research-first mindset, fusing traditional capital markets research practices with expertise in environmental sciences and technology. We think creatively about climate problems and build solutions the whole market can use.

We have developed extensive analytical frameworks to rate carbon credits on their likelihood of delivering on their promised carbon emissions avoidance or removal. These analytical frameworks are deeply sector-nuanced and bring to the fore our extensive experience in rating hundreds of projects across dozens of sub-sectors.

Our ratings team is made up of over 70 analysts who create the models and perform the analysis for every rating. They combine expertise in climatic and Earth sciences, in disciplines from remote sensing, forest ecology, ecosystem disturbance, peatlands, and blue carbon through to statistics, machine learning, artificial intelligence, sell-side financial research, industrial engineering, and public policy. They hold over 20 PhDs, have published over 200 academic papers, and have been cited over 10,000 times in peer-reviewed scientific academic literature.

The market's number-one risk metric, our ratings are available on our [website](#) and dozens of marketplaces, alongside extensive analysis and research on our BeZero Carbon Markets [platform](#).

What is the Engineered Carbon Removals sector group?

BeZero Carbon's Engineered Carbon Removals methodology is a tool for users of the BeZero Carbon Rating to better understand our multifaceted approach to ratings in this sector group. We present select case studies to showcase the various parameters through which we assess risk to the carbon quality of credits and show how the sector group relates to our overarching ratings methodology.

Engineered Carbon Removals projects issue removal credits by implementing technical solutions that enhance or facilitate the removal of greenhouse gases from the atmosphere. These projects store atmospheric carbon in new carbon sinks via biological or geochemical processes. In the [BeZero Sector Classification system](#), Engineered Carbon Removals is one of six sector groups and contains four sectors and eight sub-sectors.

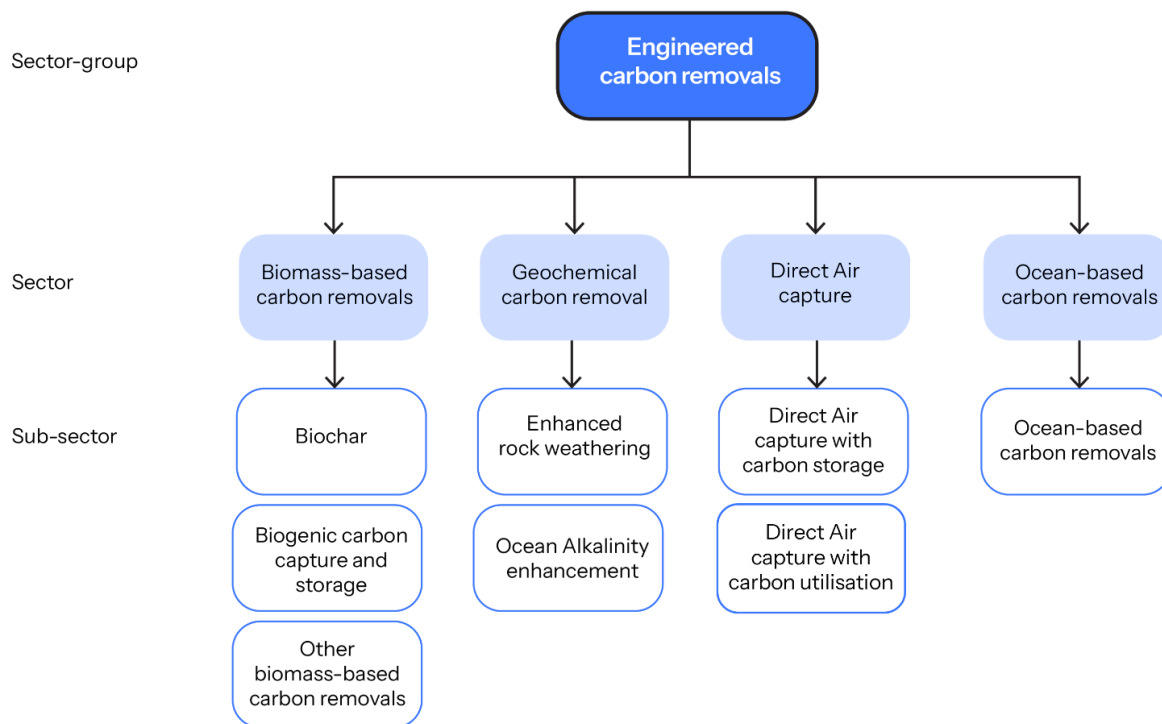


Figure 1. The Engineered Carbon Removals sector group within BeZero’s sector classification system.

BeZero has several [Insights](#) specific to this sector group, designed to better understand and inform on this nascent and largely future-facing sector. This includes the [Scalability Assessment for Carbon Removal](#), which assesses the barriers to scalability of five methods under a first-of-its-kind assessment framework. As well, the [Durability Assessment for Carbon Removal](#), which assesses the durability of seven carbon storage methods, a characteristic that sets this sector group apart from the rest of the VCM.

This methodology will provide an initial overview of the rating approach for the Engineered Carbon Removals sector group, using mostly top-down examples.

Market context

The Engineered Carbon Removals sector group is nascent. As presented in Figure 2, the majority of annual carbon removal is achieved through nature-based removal methods such as [Afforestation, Reforestation, and Restoration \(ARR\)](#), with Engineered Carbon Removals accounting for just 0.06% of total annual removal. In achieving global climate targets, this sector group will likely scale in its issuance and delivery.

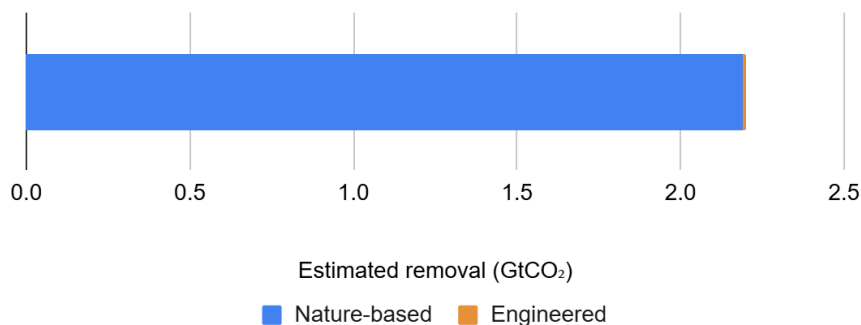


Figure 2a

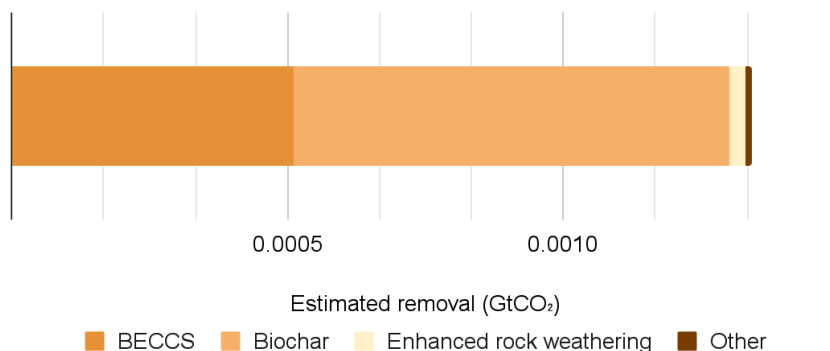


Figure 2b

Figure 2. Figure 2a shows estimated annual carbon removal in GtCO₂ from 2023. The majority of carbon removal is currently met through nature-based solutions (blue) with only 0.06% met by engineered carbon removal methods (orange). Figure 2b shows the estimated annual removal by engineered carbon removal method in GtCO₂ in 2023. Source: State of CDR 2nd edition, 2024.

As of January 2025, approximately 12.5 million credits have been transacted within this sector group. Of this, 550 thousand credits are considered delivered, ex post credits – approximately 4.4%. At 95.6%, the majority of transactions in this sector are for ex ante credits, those that are purchased pre-issuance. According to CDR.fyi, 1.2 million tonnes of biochar have been sold, compared to 7.5 million tonnes for BECCS and 2.1 million tonnes for the more costly DAC. From the ex post credits sold and issued, 85% are attributed to biochar projects.

Engineered Carbon Removal in VCM

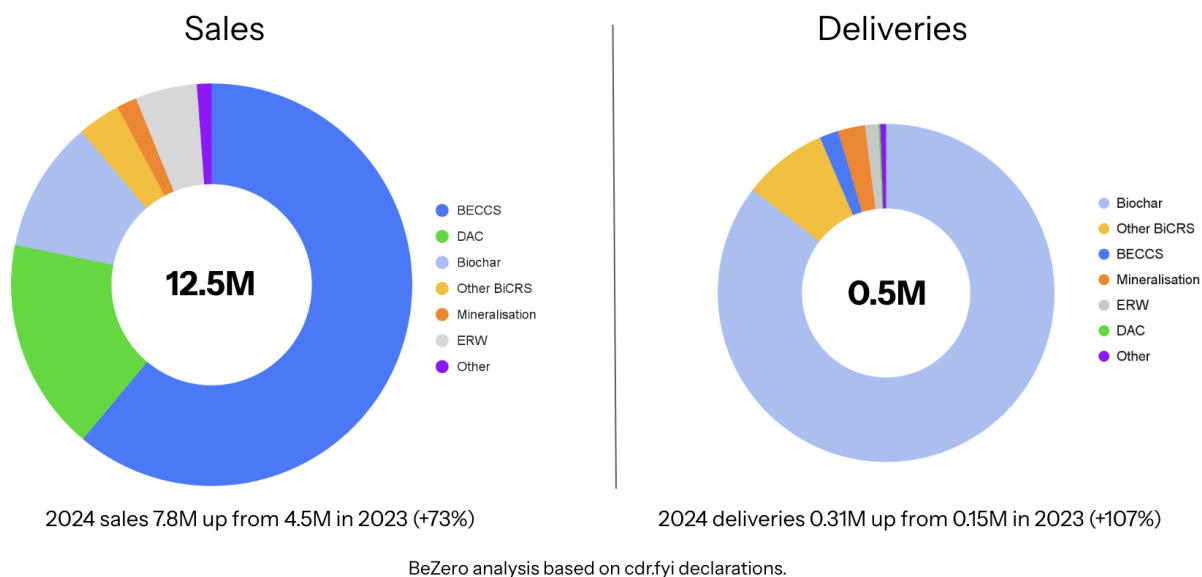


Figure 3. Volume and composition of purchased versus delivered carbon removal credits. Both sales and deliveries have grown considerably in 2024, +73% and +107% respectively. Data source: CDR.fyi.

As of January 2025, BeZero Carbon has rated 5 ex post Engineered Carbon Removals projects, and more ex ante projects as most market activity is happening pre-issuance. For projects to qualify for a BeZero Carbon Rating, they must meet our primary [qualifying criteria](#).

Comparatively, in the wider VCM, approximately 2 billion credits have been issued, of which 283 million credits have been issued in 2024 alone. The total issuance by engineered carbon removal sector group represents less than 1% of total VCM issuance by volume.

There are 34 methodologies for the Engineered Carbon Removals sector group, as of January 2025. Isometric has the greatest number of methodologies by any standard body, 8 methodologies. Puro.earth has supplied the most credits, with the most frequently used methodology being its biochar methodology with 52 projects.

Table 1. Standards bodies with applicable methodologies for Engineered Carbon Removals projects, as of January 2025. Some standards bodies cover avoidance and removal in one methodology.

Standards bodies	Number of CDR methodologies (of which cover both avoidance and removal)
Carbon Standards International	3
Isometric	8
Puro.earth	5
Riverse Carbon	2
Verified Carbon Standard (VCS)	3 (2)
Gold Standard	2 (1)
American Carbon Registry	1 (1)
Running Tide (ceased operations)	2
Open Natural Carbon Removal Accounting (ONCRA)	2
Project-specific	6
Total	34 (4)

The Engineered Carbon Removals sector is financially distinct compared to other sectors in the VCM. These nascent technologies have higher costs attributed to being early-stage. Subsequently, credit pricing is currently orders of magnitude greater than alternative sectors of the VCM - as demonstrated in Figure 4.

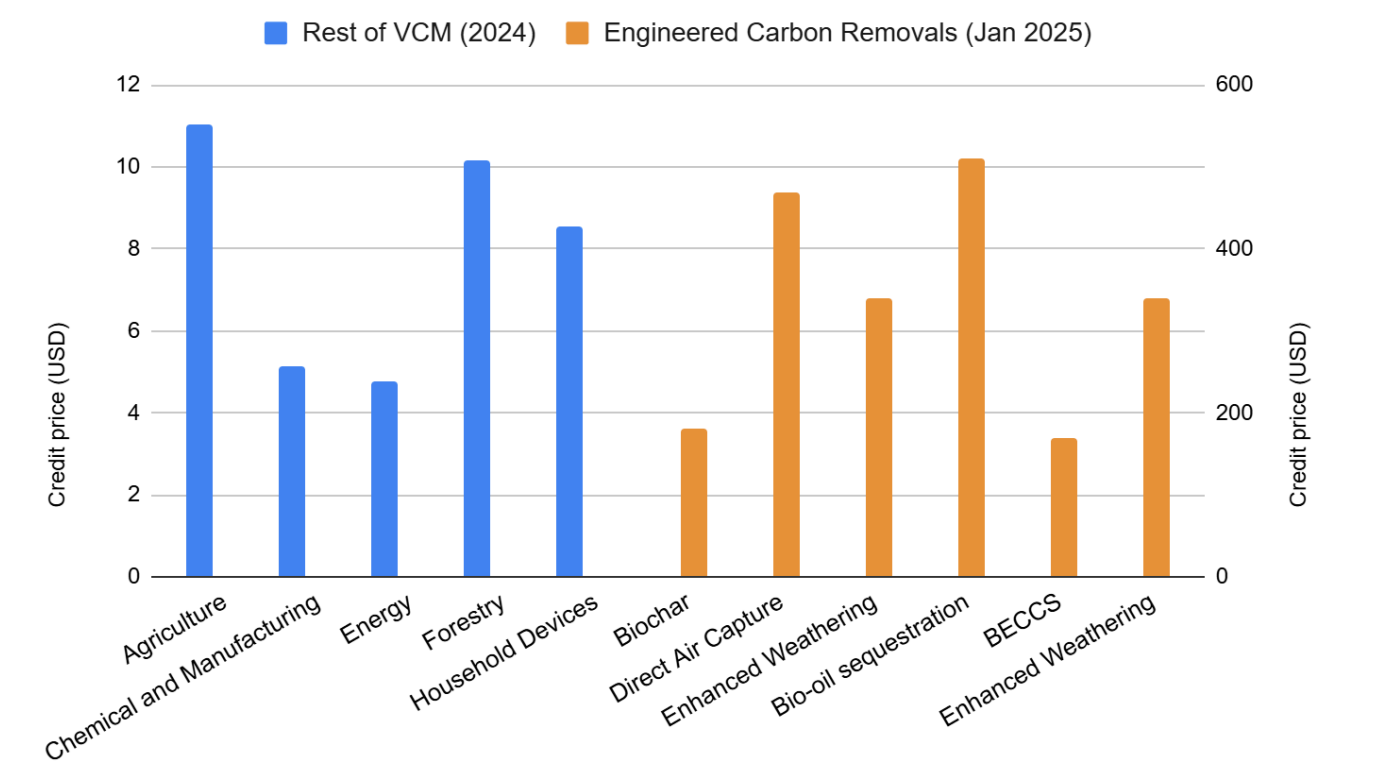


Figure 4. The Engineered Carbon Removal sector (orange) has a much higher credit price than other sectors in the voluntary carbon market (blue). Data from 2021 to today finds that credit prices in this sector are orders of magnitude greater than the rest of the VCM. Source: cdr.fyi, 2025; BeZero Carbon analysis, 2024.

BeZero Carbon Rating analytical framework and process for Engineered Carbon Removals

We have designed an approach to assessing the carbon efficacy risk for issued carbon credits. This framework is applicable to any project type in any sector accredited by any standards body, and leverages a blend of qualitative and quantitative factors; financial, environmental, and policy assessment techniques; and primary and secondary data sources.

Our [BeZero Carbon Rating Methodology](#) details our risk based assessment for understanding and interrogating carbon credit performance of any type, in any sector and country.

This document provides an extension to our overarching rating methodology. It presents BeZero's detailed analytical framework and process for Engineered Carbon Removals projects. Using select case studies, it showcases the various parameters through which we assess risk to the carbon quality of these credits, and how this sub-sector relates back to our overarching ratings methodology.

Holistic assessment

The assessment of a carbon credit's efficacy includes a detailed, project-specific, bottom-up, and top-down analysis, to provide a comprehensive assessment of risk. Further details on the risk factor definitions, the rating process, and the overall risk assessment can be found in our [BeZero Carbon Rating Methodology](#). For pre-issuance ratings please consult the BeZero Carbon Ex ante Rating Methodology.

Important elements of the BeZero Carbon rating process that are applicable to all project types are described in detail in these documents (but not repeated in this document). These include:

- The steps in the rating process
- The project governance assessment
- Project boundary auditing
- Assessing the risks of double counting
- The integrity of reported issuance
- Assessing rating eligibility (whether projects meet our three primary qualifying criteria)
- The rating assignment process
- The role of the BeZero Carbon Rating Committee
- Ongoing monitoring and 'rating watch' process

Sector and country analysis

Our assessments are based on all available project documentation in combination with our in-house models, frameworks, and databases. These include geospatial and Earth observation evidence and techniques where relevant, and a curated database from peer-reviewed literature, industry research, and third-party datasets totalling more than 12,000 sources as of January 2025.

Standards and methodology screening

Our analytical approach evaluates the rules of each standards body and each methodology on an individual basis. This screening includes an assessment of methodology development and consultation (for an overview of why this is important, see our [Insight on carbon market methodologies](#)). Further to this, we consider all deviations from methodologies. For Engineered Carbon Removals, this may include minimum requirements for the maintenance and calibration of monitoring equipment, deductions due to possible project emissions or embodied emissions, and various crediting lifetimes.

We also assess the renewal of project baselines to estimate the effectiveness and integrity of Engineered Carbon Removals projects where relevant. While some methodologies require projects to renew baselines at various defined intervals, this may not be adhered to, or the intervals may not be frequent enough to capture factors that can affect the baseline’s validity.

As outlined in the Introduction, there are 34 verified methodologies for the Engineered Carbon Removals sector. There are large variations between projects that could use the same methodology, and some sub-sectors are yet to have an applicable methodology available in the market. Further, some methodologies for a sub-sector are not applicable to specific projects or regional elements.

As the sector group develops and more projects apply these methodologies in practice, the methodological uncertainties may likely decrease. However, in its nascent state, there are inherent uncertainties surrounding methodologies that inform our analysis.

Under the top-down analytical lens of evaluating risks associated with methodologies and whether projects conform to them, we present key parameters in the assessment of Engineered Carbon Removals and the methodologies available to date in Table 2.

Table 2. Key parameters of methodology screening for Engineered Carbon Removals.

Top-down parameter	Additionality	Carbon accounting	Permanence
Methodology	Assessment of additionality	Baseline assumptions	Commitment periods
	Financial analysis	Monitoring practices	Use of buffer allocations
		Sources of project or life cycle emissions	
		Ex ante versus ex post issuance	
		Feedstock sourcing	
		Approach to leakage discounts	

Monitoring of registry operations and credit tracking also form part of our analytical process. We screen registry and standards body rules and processes, and take account of any strengths or weaknesses in our assessment of relevant risk factors.

Project- and vintage-specific analysis

Our bottom-up assessment considers all publicly-available project documentation and data, including that provided by the standards body, registry, or project developer, and any information from third-party sources, and data sourced using our internal models, notably including proprietary geospatial and Earth observation evidence and techniques where relevant.

Vintage-level assessments are made on two fronts:

- 1. Project reporting and crediting:** Our analysis ensures that, across each ratable vintage for a project, we identify whether projects correctly issued credits towards the market and buffer pool and that where credits are transferred, vintage labelling correctly maps onto cancellation certificates.
- 2. Risk factor assessments:** For each of our carbon efficacy risk factors, our analysis spans each ratable vintage of a project. This allows us to incorporate changes in project boundaries, baselines, issuance and buffer contributions over time. It also enables a dynamic process for assessing the role of policy, changes in the investment landscape for removals, and other exogenous factors that can influence broader trends in the Engineered Carbon Removals sector.

Example 1. Rating assessments require vintage-level analysis

To illustrate why ratings require a vintage-level analysis, and how we approach this, we draw on an example of one European-based biochar project.

The project issues credits under the standard body Puro.earth, and has been doing so since January 2021. This project's issuance and retirements have been reported by the standard body for this period (2021 to present). However, the project has also issued credits under an alternate standard body, Carbon Standards International, for the period from June 2018 to January 2021, using a different methodology to calculate credit issuance.

Puro.earth is a new standard body that did not begin reporting project transactions until 2021. This biochar project chose to transition to Puro.earth as early as it could. The alternate registry does not provide granular detail on credit issuance or project documentation, therefore, issuance under this standard body is not currently eligible for a BeZero rating.

Credit quantification and reporting practices vary between standard bodies for biochar. Therefore, for this project, vintage-level analysis was required in this assessment of risk.

Carbon efficacy assessment

Additionality

‘The risk that a credit purchased and retired does not lead to a tonne of CO₂e being avoided or sequestered that would not have otherwise happened.’

Our view on a project's additionality can fundamentally limit the rating we assign to credits issued by that project. This functions in two major ways:

1. Additionality is fundamental to any project being eligible for a rating. Projects that do not provide evidence to demonstrate additionality may not pass our [rating eligibility](#) process (discussed above).
2. Poor additionality constrains the overall carbon rating.

Whilst the foundational questions that underpin a BeZero Carbon Rating converge at the risk factor level, the inputs that are used to answer these questions show increasing specificity at the sub-sector and project levels.

In Table 3, we highlight sub-components of our assessment of additionality for Engineered Carbon Removals projects. In the sections that follow, we detail the key parameters that factor into each level of risk assessment and illustrate with case studies where relevant.

Table 3. Sub-components of additionality assessed.

Risk factor	Risk driver category	Risk driver	Risk sub-driver
Additionality	Activity analysis	Common practice	
		Alternatives to the proposed project	
	Financial analysis	Investment analysis	Carbon revenues
		Prior consideration of carbon finance	
		Benefit sharing	
		Barrier analysis	Financial barriers
			Non-financial barriers
	Legal and policy	Land and carbon rights	
		Policy support	Regulatory surplus
			Ownership
			Government effectiveness
			National climate and green investment landscape
			Major policies
			Nationally Determined Contributions
			National classification

Activity analysis

Common practice

Our assessment of common practice examines the pervasiveness of the project activities or technology applied by the project within the region and sub-sector with and without access to carbon finance. Projects tend to assess common practice based solely on the number of similar carbon projects operating in the region or country. We think it is important to also assess similar activities in the region or country that have been implemented without carbon finance. This helps to inform additionality as we can evaluate the potential for project activities or outcomes to have occurred either in a business-as-usual scenario or as a result of similar initiatives.

We leverage our internal markets database to assess the prevalence of all other voluntary carbon market projects in the area. Furthermore, exhaustive literature reviews are also conducted to unearth any programs or interventions conducting similar activities that may have otherwise been overlooked.

For Engineered Carbon Removals, we often find there is a lack of common practice for these technologies. The sector is nascent with a limited activity track record and limited global penetration. However, when looking closer, we find some technologies have greater common practice in some regions compared to others. For example, biochar has greater market pervasiveness in North America and some European countries than in other continents or regions. This regional aspect feeds into our assessment of common practice.

Alternatives to the proposed project

Another key aspect of additionality assessments is understanding the likelihood and extent of alternative scenarios that may occur in the absence of a project. For Engineered Carbon Removal projects, these may include the implementation of the project activity in the absence of carbon finance, the use of a more common production method or technology to achieve the same outcome, or the implementation of the project at a scale different from that proposed, among other examples. It is important to examine the credibility of each scenario to assess whether one is more likely to occur, as this necessarily influences the baseline.

This component of our analysis is interlinked with our assessments of common practice and barriers faced by the project. However, our analysis of alternative scenarios requires an even more nuanced approach, as there may be specific characteristics within the project area that make regional or even local common practices less favourable.

Example 2. BeZero evaluates the likelihood of alternative scenarios in a European biochar project

To demonstrate how we assess the likelihood of scenarios we can look at a European biochar project. This project claims removal credits based on the net carbon removed and stored in the biochar. It is a small-scale project, distributing biochar locally for application in agriculture. It uses waste wood feedstocks sourced locally.

The project claims that in alternative scenarios the waste biomass used as a feedstock would likely experience one of the following scenarios:

1. the biomass would decay,
2. the biomass would be burned to clear the waste wood, or
3. the biomass would be used in wood in construction carbon projects locally.

In our assessment, we analysed the likelihood of all three scenarios. With limited information provided on the source of the waste wood, we assessed regional data on waste biomass supplies and their life

cycles. From this analysis, we found evidence to suggest that in the absence of the project, alternative scenarios (1) and (2) are likely.

In scenario (3), the use of this biomass in an alternative carbon project would indicate alternatives to this biochar project. (It would also indicate the baseline scenario would not be 0 tCO₂e which we would address in our assessment of [the baseline scenario](#)). In this scenario, the alternative would be storage of the biomass in the construction industry, which indicates an existing removal and storage of the CO₂.

Despite identifying this scenario, the project did not disclose information to support or dispute its likelihood. We reviewed the scale of the wood in the construction industry and the penetration of such carbon projects locally as well as nationally. Our analysis found limited risk to the waste wood's use in an existing carbon project in the area, as there is no evidence of such projects regionally. Moreover, the wood in the construction industry is relatively small-scale regionally. Further, the biochar project's small scale incurs a small demand for waste biomass feedstocks, limiting the risk associated with competing demand. In our view, we found scenario (3) as unlikely and, in this case, creating low risk to the project.

Our review of relevant literature found that the overall risk to additionality is limited. However, risk is introduced due to this claim in the project documents.

Financial analysis

Investment analysis

Within the sector, the demonstration of financial additionality through investment analyses is currently uncommon. However, where data are available, our analyses evaluate the sensitivity of investment analyses, the appropriateness of any benchmarks used, the inputs to the projected internal rates of return (IRRs), and any applied sensitivity analyses. It is crucial to evaluate these inputs as the calculation of an IRR can be sensitive to certain key factors and may entail a relatively high level of associated uncertainty. Consequently, even minor variations in the key inputs can result in an artificially inflated or underestimated IRR that does not accurately reflect the project.

Our analysis therefore seeks to consider the investment barriers that may be faced by project activities by corroborating the validity of the data underlying a project's additionality assessment using independent data sources, industry data, peer-reviewed research, and in-house expertise. Factors assessed include financial inputs and trends of investment and capital flow before the project's implementation, amongst others.

For Engineered Carbon Removals, one investment barrier currently faced by projects is high capital expenditures (CAPEX). These costs are attributed to the nascent nature of these early-stage technologies, where there are high costs associated with new infrastructure and complex technology value chains. These costs inform the high credit prices (see Figure 4). They are unlikely to decrease significantly in the near term due to the nascent nature of the sector and act as a financial barrier for project activities. Compared to nature-based removals projects, such as afforestation, these CAPEX costs are orders of magnitude greater (see Figure 5). CAPEX costs are informed by project-specific information (for example project size, annual removal capacity, location, technology choices, or the policy environment) which creates large CAPEX ranges for different projects of the same technology type. However, in general technologies in this sector have high CAPEX costs which in itself likely drives higher additionality.

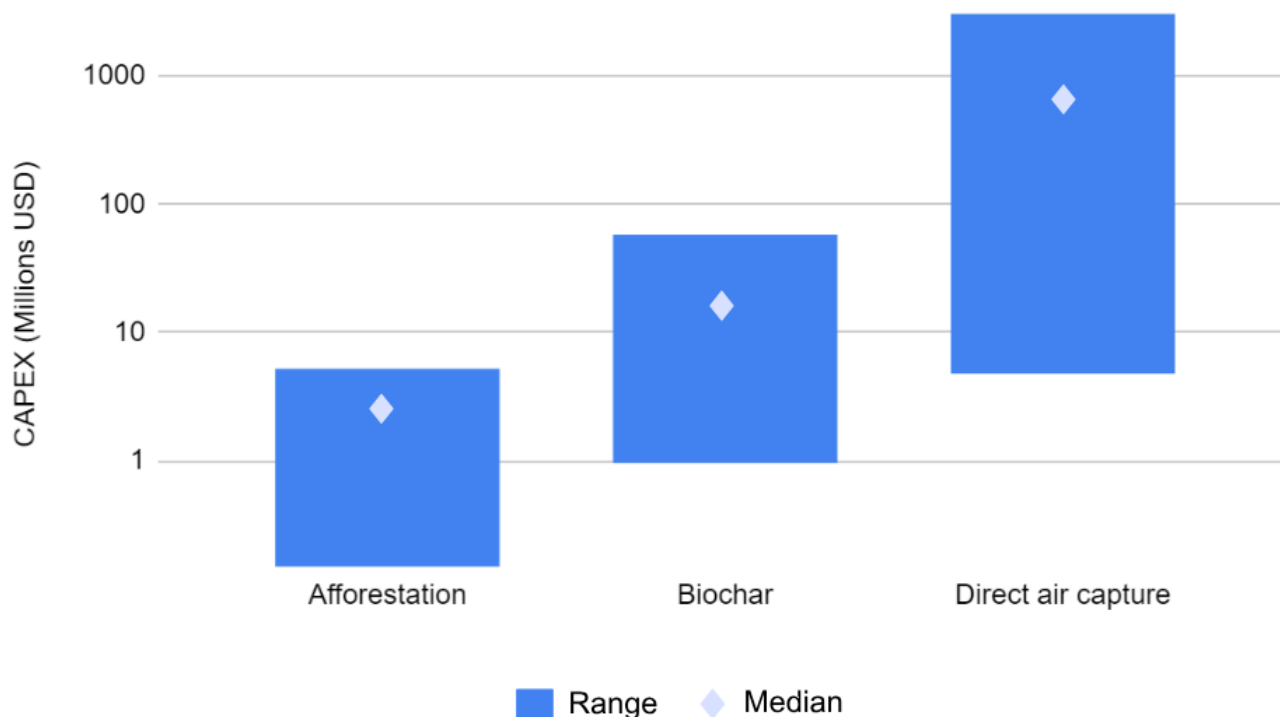


Figure 5. Estimated capital expenditures (CAPEX) for afforestation, biochar, and direct air capture projects on a logarithmic scale. In general, CAPEX costs are greater for this sector group which informs our assessment of additionality.

Carbon revenues

For Engineered Carbon Removal projects, carbon finance can serve as a supplementary or primary revenue source. For instance, in projects involving biochar, carbon finance may supplement primary revenues, while in projects like direct air capture with carbon storage (DACCS), carbon finance could form the core revenue stream. In cases where carbon finance is not the primary revenue source, other revenue streams can support project operations. Our assessment focuses on evaluating the extent and significance of these alternative revenue sources to ensure that project interventions are at least partially dependent on carbon finance and are unlikely to have otherwise occurred. For example, projects utilising pyrolysis or combustion may generate bioenergy and electricity that can be sold to the grid.

Our revenue analysis compares the contribution of multiple revenue and finance streams available to a project. This evaluation informs our opinion on additionality, as we assess whether project interventions are dependent on carbon finance and unlikely to have occurred without it. By leveraging national, regional, and project-specific data sources to identify the presence and magnitude of alternative revenue streams, we evaluate the significance of carbon finance for project implementation.

Example 3. Additionality risk increases when carbon finance is not necessary for project activities

To illustrate our assessment of revenue analysis, we consider an example of a theoretical DAC and carbon in concrete project*. The project seeks to store atmospheric CO₂ in concrete blocks.

The project listed high upfront capital costs, which we validated as likely against peer-reviewed literature reporting. These upfront costs were identified as met through equity financing and philanthropic funds. Now in its operation, the project is gaining carbon finance through credit sales, identifying carbon finance as the main source of revenue.

Analysing the revenue streams, we find inbound revenue from both the sale of concrete as a product and the sale of carbon credits. We find the main source of revenue for this project is from the sale of the concrete, at a share of 80% from concrete to 20% from carbon credits. Our analysis found that the project’s claim that the majority of revenue comes from carbon finance is likely overstated given the concrete sales. However, despite this alternative revenue, carbon finance still makes up a significant portion of the project’s total revenues. Therefore, we find a likely additionality from this revenue analysis.

In this case, the revenue from carbon finance, while not the majority revenue for this project, demonstrates a likely additional revenue for project operations. Overall, we find a low risk to additionality.

**NOTE. The carbon in concrete rating on the BeZero Carbon Markets platform (as of January 2024) is not an Engineered Carbon Removals project as the source of the CO₂ feedstocks is not atmospheric but fossil-based.*

Barrier analysis

For Engineered Carbon Removals projects, barriers arising from opportunity costs, inadequate upfront capital, technical capacity, and scientific maturity, may inhibit activities from occurring and support a project’s additionality. These barriers are often unique to the project’s location and circumstance.

To assess the presence and scale of these barriers, our analysis for Engineered Carbon Removals projects is driven by academic literature, industry reports, technical specifications, and market data. Not only does our assessment consider the existence of barriers but also the role of carbon finance in alleviating them. The latter aspect is pivotal as some barriers may be overcome by means other than carbon finance. Barriers can span several different categories, some of these are tabulated below.

Table 4. An indication of the barriers that may affect the implementation of carbon projects.

Investment barriers	While potential cost savings may incentivise the implementation of project activities, this may be countered by the lack of access to upfront capital. Securing upfront investment may be more difficult for activities in developing or low-income countries where the financing mechanisms that exist in more developed markets are generally seen to carry increased risk. As part of the larger assessment of additionality, we consider how having access to carbon finance, an external revenue stream, can aid a project in accessing the capital required to be implemented.
Technical barriers	Technological challenges may similarly prevent a project's implementation where insufficient investment and financing exist. An example may be access to technical expertise, which a formalised structure such as the carbon market could facilitate, or investment in new technologies that are not in widespread commercial operation. The nascent nature of this sector group heightens the extent of technical barriers, as while the demand is small, the supply of expertise or technologies is equally small.
Scientific barriers	Scientific challenges can arise if there are opposing views on implementing these emerging technologies. The scientific literature available for this sector group is rapidly expanding. As such, new insights can slow progress in this sector if it proposes new perspectives. While scientific certainty in these removal activities is necessary, at times, the scientific immaturity of this sector group means new insights can stall developments until this new research is fully investigated.

National barriers

We find that barriers to investment in emergent technologies can often exist at a national level, which can help substantiate claims of additionality. To assess these, BeZero conducts in-depth literature reviews considering the context of a country's industrial environment whilst also relying on other independent forms of national classification; an example being the United Nations' (UN's) least developed countries (LDCs). As defined by the UN, these are low-income countries confronting severe structural impediments to sustainable development.

Investment, scientific, and national barriers are evaluated in more detail in other sections of this methodology. Technical barriers are a critical area of analysis for Engineered Carbon Removals projects, as these projects tend to be technically complex, at times unproven, and require expensive, specialist infrastructure. Our assessment focuses on whether the project activity faces technical challenges that pose barriers to successful implementation, and on the way that increased financial resources may be able to overcome these technical challenges, such as through the hiring of specialist staff, investment into research and development, or the purchase of additional equipment.

Technological maturity

Engineered Carbon Removals is the most nascent sector group in the carbon market. In its Sixth Assessment Report, the Intergovernmental Panel on Climate Change (IPCC) scored methods within this sector group as well as nature-based removals with a Technology Readiness Level (TRL). TRL is a nine-point scale from 1-9 that indicates a technology's maturity. Technologies on the lower end of the scale, with a TRL of 1-3, are predominantly research-driven, from basic principles to proof of concepts. Technologies scoring a 4-6 represent those in development, from laboratory scale to pilot tests. Those with a score of 7-9 indicate technologies that are in their demonstration, from prototypes to proven technologies that have commercial viability.

Compared to nature-based removal methods, technologies in this sector group consistently score lower on the TRL scale. This indicates that they face barriers, which in our view, increases the additionality. The IPCC assessment of additionality is in Figure 6.

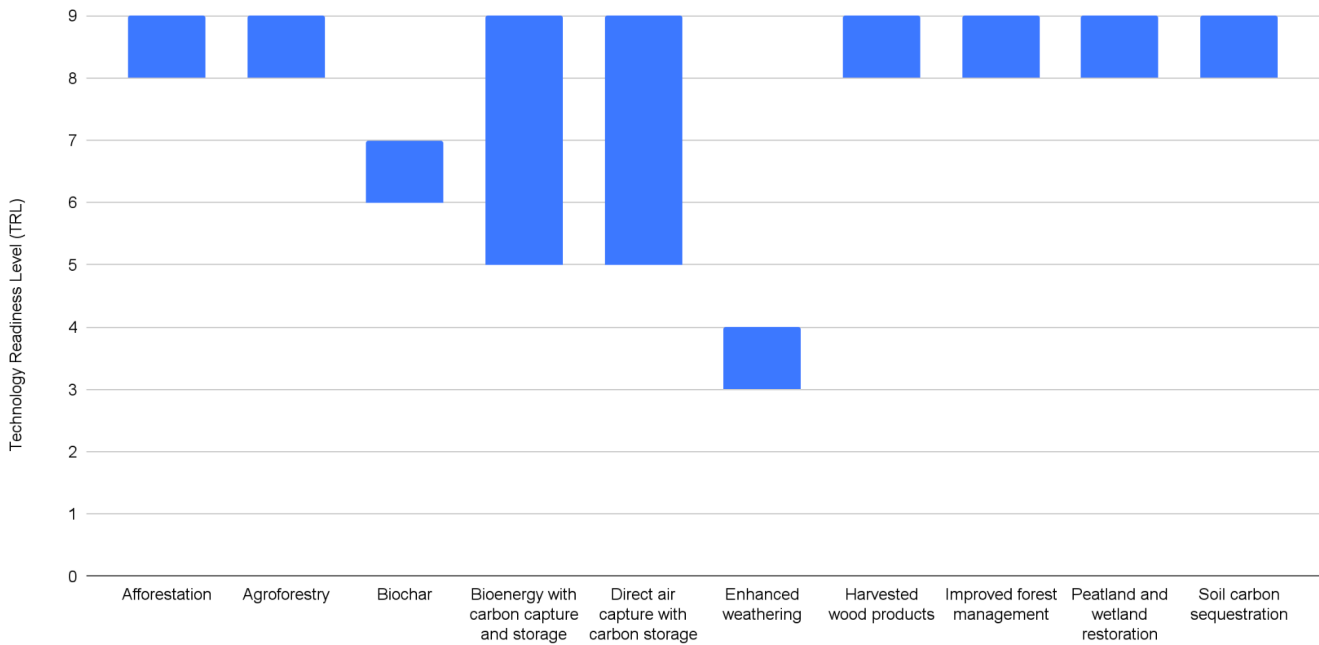


Figure 6. The technological status of different carbon removal methods, as reported in the IPCC's Sixth Assessment Report - Climate Change 2022 Working Group III. Technology readiness level is a measure of the maturity of the method. Scores range from 1 (basic principles defined) to 9 (proven in an operational environment).

This IPCC assessment in TRL has been completed at the method level. Between projects and developers of the same method, there can be variations in TRL. Therefore, it is important to combine project-level details in our assessment of technological maturity.

Benefit-sharing

The establishment of a benefit-sharing scheme, which can improve support for project activities among local communities and stakeholders, often relies on external finance. Appropriate benefit-sharing can provide long-term revenues and better equip project stakeholders to protect the area over the project lifetime and may provide an additional incentive to do so.

To conduct these assessments, BeZero thoroughly examines project documentation and pertinent literature to determine the presence of benefit-sharing mechanisms. We evaluate whether the structure of these mechanisms is well-defined and effectively communicated to local stakeholders. Our analysis encompasses several key factors, including the identification of stakeholder grievances, the establishment of consultation channels, the adherence to principles of free, prior, and informed consent; and the implementation of forward payments where applicable.

Legal and policy

Land and carbon rights

This sub-component of additionality evaluates the rights to access and implement project activities. This includes the legality of land ownership, i.e. rights within the described project area for the duration of the commitment period. We evaluate the role of carbon finance in acquiring easements or land tenure for communities. Further, we evaluate the risk to project effectiveness associated with ambiguous land rights schemes and the relative stability of national land tenure structures. For Engineered Carbon Removals projects that inject into the subsurface, we equally evaluate ownership of any sites of injection.

This sub-component also includes ownership of credit issuance. For Engineered Carbon Removals, project activities can be complex, with multiple project partners key to delivering carbon removal. For example, in DACCS projects both a capture and storage provider are necessary with a single for the delivery of removal credits. In our assessment, we analyse whether there is clear ownership of the credits issued by the project.

Policy support

Our assessment of policy considers the risk to carbon efficacy from the policy and political environment in which the project operates. This incorporates an assessment of the extent to which the policy backdrop is supportive of project activities and will consider the existence, relevance, and effectiveness of such policies. A very supportive policy environment could somewhat undermine a project's carbon efficacy. Conversely, successfully operating in an unsupportive policy environment may reinforce carbon efficacy.

Our focus is on the wider policy backdrop, beyond policy catalysts that may have already been considered relevant in our analysis of other risk factors. For example, if project activities are materially driven by policy and regulation, we would factor that into our analysis of common practice, the project counterfactual, and additionality.

Regulatory surplus

Our assessment of regulatory surplus interrogates whether there are existing laws, regulations, statutes, legal rulings, or other regulatory frameworks that support the implementation of project activities (either directly or indirectly). This could include access to regulated markets such as the Emissions Trading Scheme (ETS), or access to direct public funding. However, these assessments are

only conducted at a single point in time and are therefore not likely to reflect the broader regulatory environment preceding the project's start date nor throughout its crediting period, should sufficient evidence be available to support this analysis. We will explore the regulatory environment that pertains to the development and entire operational period of the project where possible. For Engineered Carbon Removal projects, this typically involves investigating regulations related to increased adoption or development of negative emissions technologies.

We identify government or external regulations that require the development and installation of Engineered Carbon Removals projects. We also identify regulations that dictate how projects in this sector group can develop, i.e., restrict certain activities unless specific requirements are met. These could be implemented as a measure to overcome uncertainties or to ensure certain standards are kept. In our view, such regulations can limit a technology's scale-up and inform a project's additionality.

The impact of this on our rating is closely related to our assessment of government effectiveness, where we analyse the implementation, enforcement, and overall effectiveness of policy regulations and initiatives. Where we find evidence of regulations that mandate project activities in combination with high levels of government effectiveness, we would consider this to increase additionality risk.

Government effectiveness

In our assessment of Engineered Carbon Removals projects, we thoroughly evaluate the effectiveness of government policies at the national, regional, and local levels. Governments hold a crucial responsibility in formulating policies, allocating resources, and coordinating actions. This assessment encompasses various aspects, such as the presence of well-defined comprehensive legislative frameworks and regulations that support the implementation of Engineered Carbon Removals projects.

Our analysis encompasses the implementation, enforcement, and overall effectiveness of policy regulations and initiatives. Additionally, we consider the allocation of financial resources, both domestically and through international cooperation, to facilitate Engineered Carbon Removals projects.

National climate and green investment landscape

For Engineered Carbon Removals projects, we first assess investment opportunities and activities related to climate change mitigation and adaptation within a specific country. This may include the presence of policies, regulations, and incentives – such as tax relief, grants, subsidies, and the market potential for removal projects in the relevant country – that support the project's activities.

Second, we identify ongoing and planned projects, their scale, scope, and location, and assess the number of alternate projects that could implement the project activities and the technologies they currently employ.

Third, we identify the potential risks associated with investments in Engineered Carbon Removals projects, such as policy and regulatory risks, market volatility, supply and demand issues, and lack of infrastructure.

Fourth, where relevant, we consider factors such as project readiness, scalability, and potential co-benefits to determine attractive investment opportunities and evaluate the financial viability, profitability, and risk-return profile of different investment options.

Major policies

As part of our assessment of additionality, we analyse any major policies or initiatives that may affect the country or region where the project is located. We identify government or external strategies that aid the development and deployment of the same or similar technologies in the relevant industry. In

our view, the existence of major policies and initiatives is likely to indicate a supportive environment for the project activities – as long as they are effectively implemented. If this is the case, we find that project activities may have been introduced without the need for carbon finance. However, in our assessment, we analyse how successful the policy or initiative has been. If the policy has not been implemented effectively, this may not increase policy risk, or could even decrease it. Our assessment of a project's policy risk includes all relevant policies: local, regional, national, and international. We will also consider how these regulations interact, where relevant.

Example 4. Policies in the United States introduce some additionality risk for direct air capture and other carbon capture projects on the carbon market

In recent years several major policies have been introduced in the United States. No method received more support through these major policies than direct air capture (DAC). Key policies include:

Bipartisan Infrastructure Investment & Jobs Act (2021). BIIJA introduced several carbon management programmes that DAC projects can utilise. This includes four regional DAC hubs (USD 3.5 billion), carbon dioxide transportation infrastructure finance and innovation program (USD 2 billion), carbon storage validation and testing (USD 2.5 billion), carbon utilisation program (USD 500 million), and underground injection control grants program for Class VI wells (USD 48 million). In total, this leads to USD 7.5 billion in funding made available to DAC project types.

Inflation Reduction Act (2022). This legislation is considered the largest single climate policy to date, with USD 369 billion in public funding for climate action. For DAC, the most notable advancement is enhancements of the 45Q tax credit. The size of the tax credit increased for DAC projects, the payment option for receiving the credit was extended, and the scope and scale of qualified facilities were broadened.

Class VI wells. Class VI well permits are required for any subsurface storage projects in the USA. Class VI wells have stringent requirements designed to ensure the safety and durability of CO₂ injection. As of December 2023, 12 Class VI wells have been permitted in the USA.

The existence of regulations incentivising the project activity in all relevant jurisdictions suggests to us that the policy environment for this project is supportive. Nevertheless, this support is relatively recent and limited in effect. This approach to policy analysis allows us to rate projects of varying scopes, including those that are international and therefore subject to a complex regulatory environment.

Nationally Determined Contribution

Assessing the Nationally Determined Contribution (NDC) of the project's host country is a factor in our top-down policy analysis for Engineered Carbon Removals projects. This refers to the commitment made by each participating country to outline its efforts in mitigating greenhouse gas emissions and adapting to the impacts of climate change.

NDCs may consider Engineered Carbon Removals to be a high national priority from an emissions perspective, suggesting that the government may wish to implement project activities (or other, similar sustainable practices) in the absence of carbon finance. In time, as countries consistently miss their NDC targets they will likely increasingly look towards the Engineered Carbon Removals sector group as a solution regardless of carbon finance. This is important for us to evaluate over a project's lifetime.

National classification

We use a country's national classification to understand the barriers to project activities that are inherent to the country in which a given project operates. These national classifications are important to inform the context in which the project operates. For example, a classification as a 'Least Developed Country (LDC)' (as defined by the United Nations) may be relevant to the risk of investment.

Typically, we would consider projects in LDCs to face less policy risk, as there would be less chance that the policy environment would lower the project's carbon effectiveness. In contrast, a high-income country would be considered to face more policy risk as it is likely that there would be alternatives to carbon finance to overcome barriers.

Carbon Accounting

‘The risk that the carbon accounting underlying a credit does not fully achieve a tonne of CO₂e avoided or removed.’

In Table 5, we highlight sub-components of our assessment of carbon accounting for Engineered Carbon Removals projects. In the sections that follow, we detail the key parameters that factor into each level of risk assessment and illustrate with case studies as relevant.

Table 5. Sub-components of our carbon accounting assessment.

Risk factor	Risk driver category	Risk driver	Risk sub-driver
Carbon accounting	Direct accounting	Baseline scenario	Validity of zero baseline claims
			Alternative feedstock scenarios
		Project scenario	Measurement and monitoring practices
			Project emissions
			Life cycle emissions
		GHG conversions	Methodological uncertainty
			Carbon accounting uncertainty
			Safeguards
	Leakage accounting	Activity displacement	
		Market leakage	
		Ecological leakage	

Direct accounting

Baseline scenario

Validity of zero baseline claims

A critical part of assessing a project’s carbon accounting risk is evaluating the baseline scenario selected by the project. The baseline of a carbon project is the amount of emissions that have previously occurred or would likely continue to occur in the absence of any mitigation strategies. This is then typically compared with the emissions that result in the project scenario, and the difference between these two values comprises the project’s emissions reductions. An appropriate baseline relies on the use of representative historical data and/or models to forecast future emission scenarios. Ideally, the baseline would also assess the likelihood that emissions reductions occur during the project’s crediting period as a result of exogenous factors, such as improving industry standards and changes in regulation, if a historical precedent exists for this.

For Engineered Carbon Removals, the baseline is often presented as 0 tCO₂e emitted and/or removed. In our ratings analysis, we assess the likelihood of this statement in light of project-specific information. For example, carbon activities could be operating in some form prior to the formalisation of a carbon project. In our assessment, we review all activity in the project area before the project activity and analyse whether this has been appropriately captured within the project’s baseline scenario. We find that projects can overlook existing nature-based removals, slow counterfactual feedstock decay rates or alternative scenarios for feedstocks used in the project, creating a risk of

over-crediting. We also find that some projects can conservatively opt for a 0 tCO₂e baseline scenario, thereby creating the potential for under-crediting. For each project, we assess the application of the chosen baseline scenario and its risk to the robustness of the carbon accounting.

Our assessment of a project’s baseline can overlap with our evaluation of a project’s additionality risk: if it appears that the project activity would likely have occurred in both the project and baseline scenarios, this may have implications for both our additionality and carbon accounting risk factors.

Alternative feedstock scenarios

Our assessment of carbon accounting risk is informed by the alternative scenarios for feedstocks used in the activity in the baseline scenario. For biomass-based carbon removals and some ocean-based carbon removals, biogenic feedstocks are required as inputs. This can range from woods to agriculture waste, or biogenic sludge. Similarly, for enhanced weathering and some ocean-based carbon removals, mineral feedstocks are required. Where we find the alternative scenarios of these feedstocks could introduce over-crediting risk, for example, if they would have otherwise stored the CO₂ without the project, we will include this in our assessment.

Project scenario

Project emissions

Our assessment of over-crediting also considers project emissions, which are emissions that occur during the implementation of the project. These must be deducted from the project’s gross issuance to increase the likelihood of accurate crediting.

For Engineered Carbon Removals projects, project emissions often result from

- The emissions associated with fossil fuel consumed on-site;
- The emissions from any electricity consumed on-site;
- The transportation on-site or to the site for the project activity;
- The embodied emissions of materials used during operations on-site.

These sources are often small in both an absolute sense and relative to avoided emissions. However, this is not universal, and external emission sources associated with a project can present a material source of emissions. Further, for Engineered Carbon Removals projects, these emissions could be significant given the current small-scale of these projects and their inherent infrastructural requirements as technological removal solutions.

Measurement and monitoring practices

The measurement and monitoring practices employed by a project are also highly relevant to its carbon accounting over-crediting risk. For Engineered Carbon Removals, measurement and monitoring practices vary between sub-sectors. Practices in closed-loop systems tend to be more reliable than those in open-loop systems. In closed-loop systems, CO₂ can be easily quantified through techniques such as flow meters to measure variables including amount, composition, concentration, and/or pressure.

Table 6. Closed versus open-loop systems for Engineered Carbon Removals.

Closed-loop system	Projects that involve a high degree of technical control or containment. This includes methods such as bioenergy with carbon capture & storage (BECCS), bio-oil sequestration, or direct air capture with carbon storage (DACCS). In these systems, there tends to be greater certainty in measurement and monitoring as the CO ₂ can be frequently measured with equipment, such as flow meters.
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Open-loop system

Projects that involve processes outside technical control or containment. This includes methods such as enhanced weathering, ocean alkalinity enhancement, or other ocean-based carbon removals. Open systems have greater uncertainty in their measurement and monitoring practices.

We find the frequency of monitoring practices to be more significant in our assessment of over-crediting risk for open-loop systems, where there is a greater risk due to reported greenhouse gas flows. Variables outside of human control can create uncertainty. Measurement and monitoring practices can limit the extent of uncertainty in credit quantification, limiting overall risk. Practices could include feedstock testing, control site monitoring, or supporting modelling.

Example 5. Independent measurement and analysis of applied feedstocks reduces carbon accounting risk

We can illustrate the impact on measurement practices by looking at enhanced weathering projects.

Enhanced weathering is a geochemical Engineered Carbon Removal method, and an open-loop system, that accelerates the natural weathering process to remove atmospheric CO₂. Alkaline rocks, either silicates or carbonates, are ground into a fine powder to increase their surface area with the powder then being spread over soil. The rocks react with atmospheric CO₂, converting it into dissolved inorganic carbon in the form of bicarbonate and carbonate ions dissolved in the soil and groundwater.

Enhanced weathering projects have several uncertainties that arise in their carbon accounting. There is no current best practice for the measurement or monitoring of these projects. Both enhanced weathering methodologies currently in the market – Carbon Standards International and Puro.earth – rely on modelled approaches to quantify removal. At the project level, these modelled approaches can be supported by measured results. In general, for this subsector, measurement and monitoring practices are highly project-specific.

To limit the risks associated with scientific uncertainty, certain measurement practices at the project level can be implemented. One practice in particular is mineralogical classification prior to the grinding process and implementation.

The alkaline rocks used as feedstock can vary widely. For example, feedstocks could be basalt, serpentine, olivine, dolomite, limestone, wollastonite, or mine tailings. Depending on which feedstock is used the chemical reactions that take place will differ. The chemical reaction informs how much carbon has been removed, and at what rate, and thus the calculated credit issuance will vary. Project developers need to ensure accurate classification of the rock feedstock to deliver accurate issuance calculations. However, research finds that the rock feedstocks being utilised in some studies are not homogeneous in their mineralogy. This has unintended consequences on chemical reactions and issuance calculations.

To limit risk, the rocks utilised can be classified and coded before their use. Independent organisations can be used to provide such assessments. By implementing this measurement practice, uncertainties can be reduced, and the overall risk of over-crediting for enhanced weathering projects can be restrained.

Life cycle emissions

Our life cycle emissions analysis delves into the upstream emissions linked to the project, encompassing the manufacturing of technological components, transport, distribution, and installation at the project site. Where feasible, we quantify these emissions and deduct them from issued removal credits. The scale of the project often influences this assessment: upstream emissions are typically more impactful for small-scale projects, as they tend to constitute a larger proportion of the project's gross issuance. In the case of Engineered Carbon Removals projects, these emissions could be significant given the current small-scale of these projects and their inherent infrastructural requirements as technological removal solutions.

Example 6. Life cycle emissions upstream can create leakage risk if unaccounted

To demonstrate how we assess life cycle emissions, we can look at a small-scale bioenergy with carbon capture & storage (BECCS) project.

In Figure 7, we outline a BECCS project that has defined its project boundary to include the bioenergy facility and CO₂ storage site. From this life cycle the project has calculated its net carbon removal from the gross carbon stored at the geological injection site and subtracted project emissions from the associated arrows.

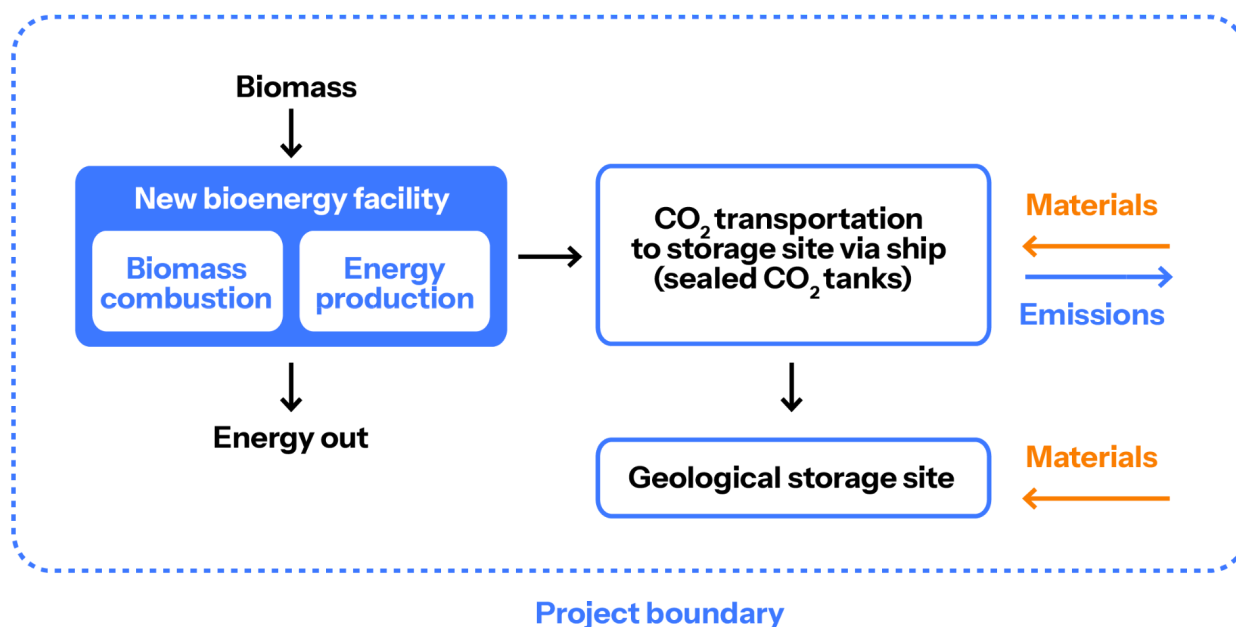


Figure 7. A life cycle of a BECCS project. The project boundary for this project includes bioenergy production and carbon storage.

This project boundary excludes several important life cycle emissions from the credit quantification and subsequent issuance. For example, the bioenergy facility has been identified as new, yet the embodied emissions associated with the construction of this facility have been excluded. Further, the upstream emissions have been excluded from the assessment.

In Figure 8, we have expanded the BECCS project's boundary to encompass cradle-to-grave emissions. Cradle-to-grave refers to the entire lifecycle from first CO₂ absorption in the biomass to final storage. In the second life cycle, the project has accounted for emissions at the new bioenergy facility in its life cycle. In addition, the expanded life cycle identifies that a new processing and

pelleting facility was built for this project. Embodied emissions and transportation emissions associated with this new facility will be included in this second life cycle assessment.

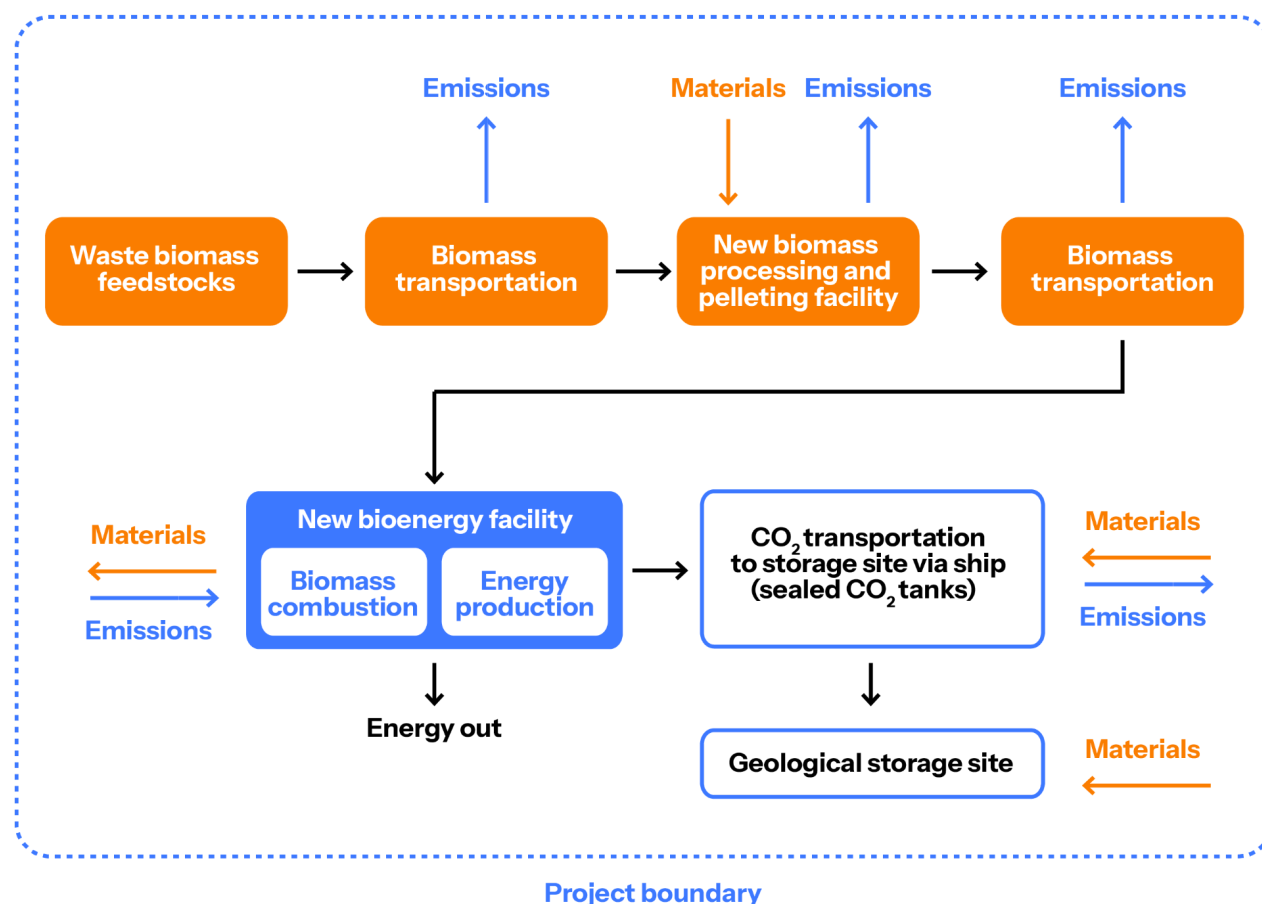


Figure 8. A life cycle of a BECCS project from cradle to grave. The project boundary has been expanded to include biomass processing and extended the number of life cycle inputs and outputs. This is a non-exhaustive life cycle.

Cradle-to-grave life cycle assessments for projects limit carbon accounting risk by including potentially significant emissions in credit calculations, such as construction emissions. Construction emissions of any new facilities will not necessarily be significant for every BECCS project. If an existing bioenergy or biomass pelleting facility is utilised, these emissions could be negligible. Similarly, if the project is particularly large-scale, these emissions could be comparatively negligible. At the project level, the inclusion of these life cycle emissions in project carbon accounting or identification and discussion of their negligible effect can mitigate leakage risk.

GHG conversions

Methodological uncertainty

As outlined in the Introduction, there are 17 verified methodologies for the Engineered Carbon Removals sector. These methodologies are specific to a subsector, with some being designed specifically for a project. The Engineered Carbon Removals methods covered by a methodology are shown in Figure 9.

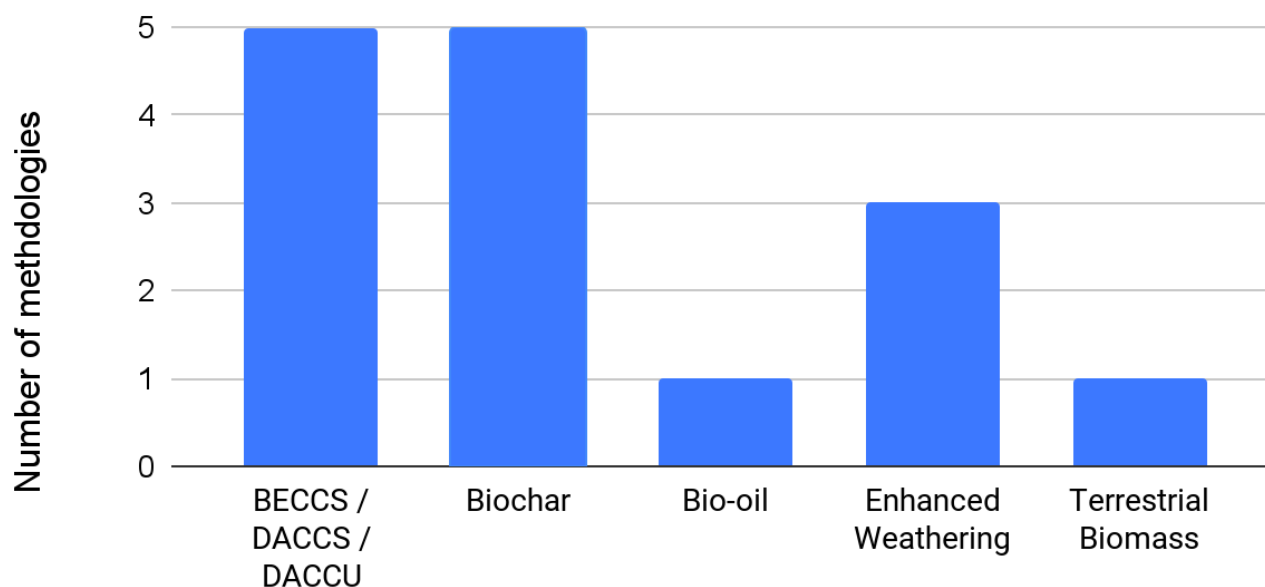


Figure 9. Number of verified methodologies by Engineered Carbon Removals method, as of January 2024.

Many of the available methodologies are being implemented for the first time. Whether designed by a standard body, or project developers opting to curate project-specific or technology-specific methodologies that are verified by an independent auditor. When methodologies are utilised for the first time, there may be no track record of their implementation. This can create risks that could temper the robustness of the accounting by excluding or undercompensating for unknown or uncertain emissions in credit calculations. Even if there is a low-risk profile for this project's GHG flows and appropriate monitoring protocols, the lack of a track record for methodology implementation creates uncertainty and risk. Without a demonstration of successful measurement in line with the monitoring plan, there will be inherent uncertainty on whether or not the activity has successfully occurred. This risk can be mitigated by safeguard measures applied by the project (see [Safeguards](#)).

Carbon accounting uncertainty

For Engineered Carbon Removals, in general, there is currently no best practice for carbon accounting. There is uncertainty in carbon accounting in many areas of this sector group. Further, credit quantification and carbon accounting can at times rely on modelled approaches rather than measurements. Modelled approaches are not necessarily of low quality because they may encompass components that are not feasibly measured. Vice versa, measured approaches are not necessarily the optimal approach, given geographical differences, distributional challenges in accessing measurement resources, and/or significant associated costs.

Carbon accounting uncertainties can introduce risk to credit quantification. These risks can be mitigated at the project level. In the near-term, some Engineered Carbon Removals methods are likely to face significant risks associated with their carbon accounting uncertainty.

Example 7. BeZero evaluates the extent of uncertainty for these novel technologies

Ocean alkalinity enhancement projects are open-loop systems that present several measurement challenges. In general, there is a lack of consensus surrounding the best approaches to carbon accounting. The uncertainty surrounding measurement is caused by multiple factors, including variations in alkalinity feedstocks, project scale, or geographical variations in climatic or ecological conditions. These challenges are exacerbated by the nature of the open-loop system where the project area is not contained.

Currently, there is no verified methodology for this technology. There are several monitoring and measurement practices identified in the literature to quantify potential carbon removal, with project developers often employing several different measured and modelled approaches congruently. This could include monitoring dilution rates and transportation through tracer methods, turbidity through microstructure profiling, residence time through risk assessment, light penetration, conductivity, and depth movement through sensors, and comparison with control experiments (this is a non-exhaustive list). Despite these techniques, a lack of demonstrated success by this method means carbon accounting uncertainties persist.

Despite this uncertainty, projects are continuing to develop due to the potential of this sub-sector as a removals solution. To limit the impact of these uncertainties, and associated carbon accounting risk, appropriate safeguard measures could be implemented by project developers. This would deliver a more conservative approach to carbon accounting to overcome some uncertainties.

Safeguards

With respect to uncertainty, safeguards represent measures employed to indicate an acknowledgement of the inherent uncertainties involved. In the case of Engineered Carbon Removals, a sector group still in its early stages, we observe that certain methodologies for projects incorporate conservative measures through the implementation of safeguards. These safeguards may encompass unallocated capacities, margins of safety, or other buffering measures. Their purpose is to mitigate the risk of over-crediting. However, in some scenarios, safeguards may inadvertently introduce the risk of under-crediting. Therefore, in our assessment of carbon accounting, we analyse any measures implemented by a project to manage and constrain uncertainty.

Leakage accounting

Activity displacement

For Engineered Carbon Removals projects, the risk of activity displacement is likely to depend on the specific project activity taking place. For projects that involve some form of feedstock harvesting, such as biomass-based carbon removals, there is an increased risk of displacement owing to conflict for biomass feedstocks. Similarly, for projects that require a land area for spreading activities, such as enhanced weathering, there is an increased risk of displacement if other activities cannot be collocated at the spreading site.

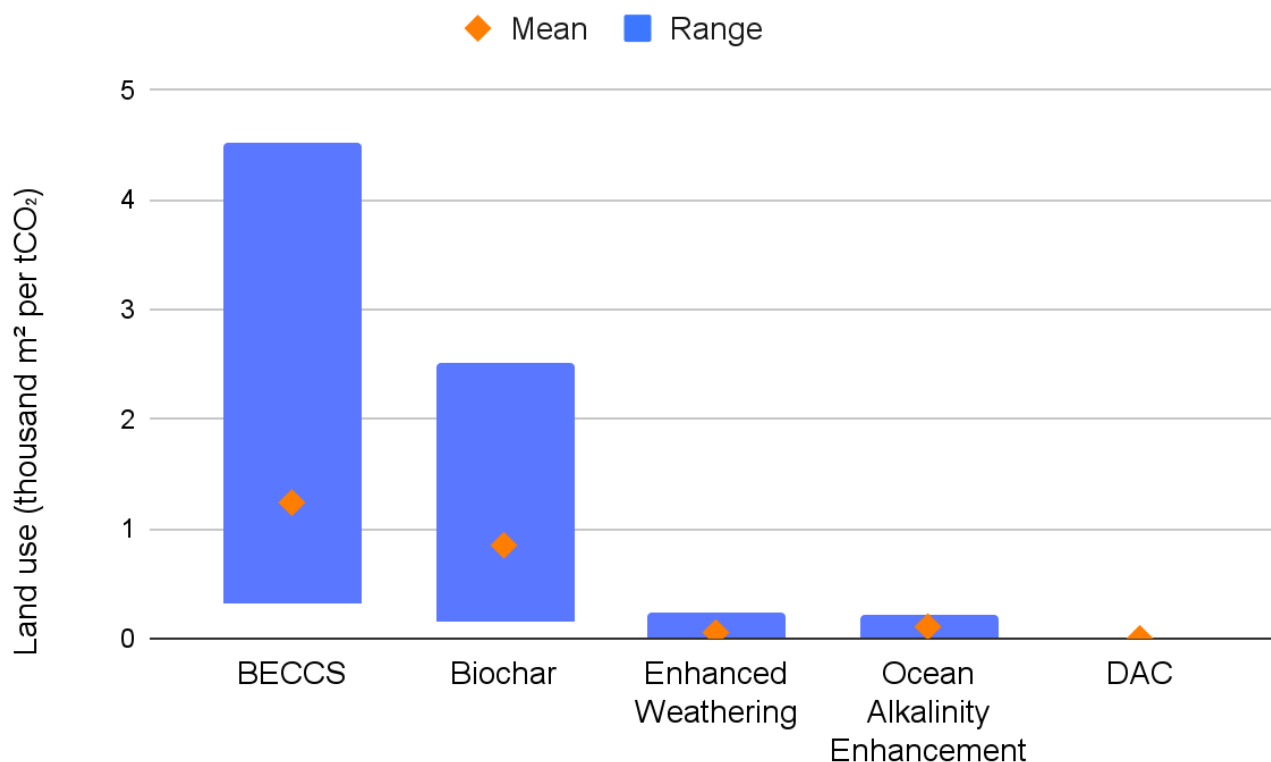


Figure 10. Land demand in metres squared as a range and an average per Engineered Carbon Removal method, from the 2023 BeZero Carbon [Scalability Assessment](#)

We evaluate the scale of the project in our assessment of activity displacement. For small-scale Engineered Carbon Removals projects, land requirements are currently likely to be minimal, limiting risk. As projects scale, this assessment will need to be revisited depending on project-specific requirements. Further, we consider whether the activity takes place at a brownfield site (an existing or abandoned site that has been previously built on) or a greenfield site (a new site with no building). The risk of activity displacement or leakage will depend on project-specific characterisation and demands.

Our assessment of carbon accounting risk due to activity displacement will also include an evaluation of transport emissions within the project area, where relevant. Transportation of materials from the harvesting site to the final storage location will be evaluated. In cases where the project activity involves significantly increased transport requirements that have not been accounted for, we may reflect this as a carbon accounting risk.

Market leakage

Market leakage can occur when a project's activities alter the supply and demand equilibrium. As a result, shifting market dynamics may lead to increased emissions due to market activities moving elsewhere and projects may need to deduct from their issuance to account for this risk. However, for this to be likely, in our view, a carbon project must influence a significant portion of the supply and/or demand for a specific supply chain. In the case of Engineered Carbon Removals projects, we find that this risk is currently negligible given the small scale of projects that are operating in the market. As projects scale and supply increases, market leakage for a project could increase.

Positive leakage

If we believe there is evidence that a project will likely prevent additional emissions beyond those caused by the core project activity, we may assess the project to display positive leakage. This may reduce a project's overall carbon accounting risk level.

For Engineered Carbon Removals projects, we find positive leakage can occur when projects credit solely for the CO₂ removal activity and not other greenhouse gas (GHG) reductions that can occur due to the project activity. Some GHG removal activity associated with Engineered Carbon Removals can be uncertain due to the extent of scientific literature published or market data that concludes in certainty. However, these positive effects could still be likely and are included in our assessment of positive leakage.

Example 8. Positive leakage accounts for uncredited greenhouse gas benefits

We can demonstrate positive leakage by looking at biochar projects.

Biochar projects most commonly issue credits based on the gross carbon crediting solely for the CO₂ stored in the biochar. For many projects issuing credits to the market, project boundaries do not encompass the application of biochar in agricultural fields. However, research suggests that biochar improves a number of soil carbon stocks. Further, the pyrolysis process produces energy that can be utilised locally, at times replacing fossil-based alternatives. These benefits are not credited by the project, likely due to a lack of certainty for the specific project without further analysis and measurement, which can be costly to prove. The carbon benefits, beyond the project boundaries are considered under our assessments of positive leakage.

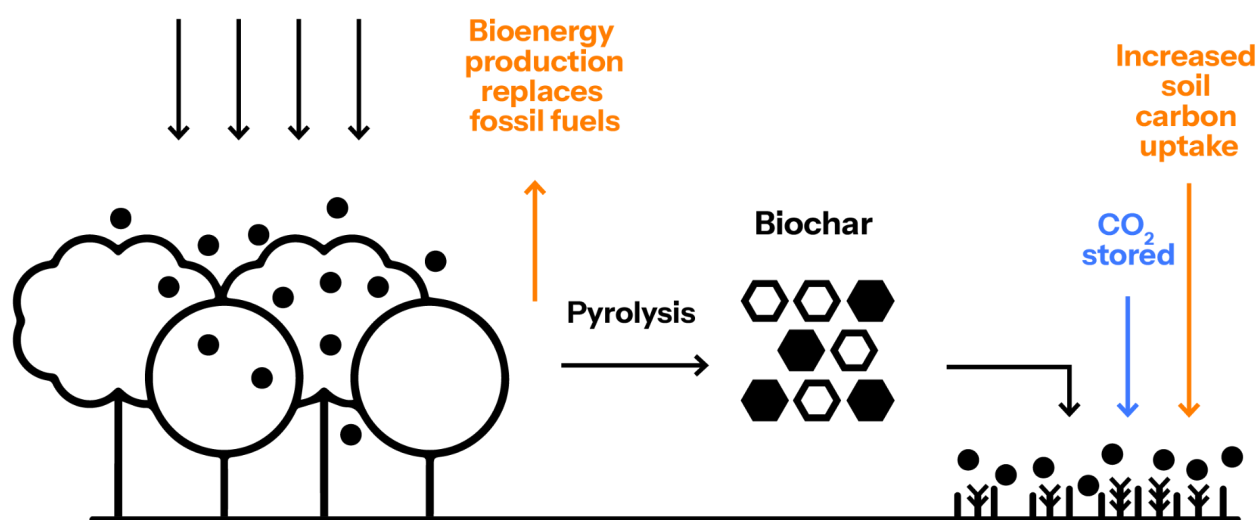


Figure 11. Diagram of a biochar project with application on soils. Activities in blue are credited for. Activities in orange are not credited for and can be considered positive leakage.

Ecological Leakage

Ecological leakage in Engineered Carbon Removals projects refers to the unintended ecological consequences that may occur as a result of implementing such projects. It refers to the displacement or shifting of environmental impacts - beyond carbon emissions - from the project area to other ecosystems or regions. These impacts can occur in both terrestrial and marine ecosystems.

Ecological leakage can be positive or negative, impacting biodiversity, habitat connectivity, and changes in ecological processes. For example, for biomass-based carbon removals or enhanced weathering projects, positive impacts could be seen on plant productivity, crop yield, water use efficiency, or reduced leaching of greenhouse gases. Meanwhile, negative impacts could arise from the use of certain feedstocks that increase ecological leaching. We undertake a full assessment of ecosystems surrounding the project area to assess the potential for ecological leakage.

Permanence

‘The risk that the carbon avoided or removed by the project will not remain so for the time committed.’

Our analysis of this risk factor considers the permanence of the contractual commitment of a given project and its proposed credits. To assess a credit’s permanence risks, it is first necessary to determine the period a credit commits to, and then whether the credit faces reversal risks within that period. Commitment periods are the duration over which sequestration or abatement activities have permanence horizons and differ from crediting periods (the timeframes during which reductions or removals are eligible for issuance as verified carbon credits).

In Table 7, we highlight sub-components of our assessment of permanence for Engineered Carbon Removals projects. In the sections that follow, we detail the key parameters that factor into each level of risk assessment and illustrate with case studies as relevant.

Table 7. Sub-components of permanence assessed.

Risk factor	Risk driver category	Risk driver	Risk sub-driver
Permanence	Anthropogenic risks	Project management and carbon rights	
	Technical risks	Carbon stock integrity	Durability
	Risk mitigation instruments	Stakeholder engagement Risk buffer and mitigation	

Anthropogenic risks

Anthropogenic risks are those introduced by human activities and that could result in the reversal of carbon sequestration. For the Engineered Carbon Removals sector, these risks can extend from the presence of legacy wells in the subsurface to the presence of biological or chemical matter. Some of these risks can be avoided at the project level. Anthropogenic risks also include any legal risks of non-permanence. Legal risk could be driven by uncertainty regarding issuance ownership and/or uncertainty regarding storage durability or insecure land tenure of the storage site over the project’s commitment period.

Example 9. BeZero analyses anthropogenic risks such as unintended acidity

To illustrate the impact of anthropogenic risks we can look to Enhanced Weathering projects.

In these projects, enhanced weathering occurs following the chemical reaction of alkaline rocks (silicate or carbonate) with atmospheric CO₂ and water. The chemical reaction that occurs is dependent on the alkaline feedstock and the conditions in the area where the activity occurs. The CO₂ is considered removed and stored following chemical reactions into either carbonate or bicarbonate minerals.

However, anthropogenic application of nitric or sulphuric acids or certain agricultural practices can increase the risk of reversal. This is because in the weathering process, if the alkaline rocks are exposed to non-carbonic strong acids, mineral weathering can occur. Mineral weathering can decrease the efficacy of CO₂ storage, with some research suggesting that it could result in the release of CO₂ into the atmosphere if carbonate rocks are used as feedstock.

Carbon rights

Carbon rights are a form of property rights that allow accredited carbon units to be owned and traded. Our assessment evaluates who owns the carbon rights for the project area and whether the ownership is likely to confer non-permanence risk. Carbon rights may be detailed in project documentation or through project-specific literature. Risk may be reduced where carbon rights are owned by the project developer. In contrast, risk may be introduced where carbon rights are owned by the government, as the project may not have the legal right to sell carbon credits.

Technical risks

Commitment period

A carbon credit signifies a contract for the avoidance or removal of 1 tCO₂e. In the context of Engineered Carbon Removals, this contract should specify a commitment period, delineating the duration over which sequestration activities are intended to maintain permanence horizons. Commitment periods differ from crediting periods which denote the time frames during which reductions or removals are eligible for issuance as verified carbon credits.

Our ratings assess the likelihood that carbon will be removed for the entirety of the project's commitment period. Therefore, in evaluating permanence, we focus on reversal risk within this timeframe, including monitoring, reporting, and verification (MRV) practices over this period.

Across the carbon market, there are no common descriptions for commitment periods, with variations between standard bodies and projects. For instance, Verified Carbon Standard (VCS) employs a maximum time horizon of up to 100 years for nature-based solutions. However, VCS do not address reversals occurring after the crediting period, rendering the commitment void after the crediting period and any buffer pool credits cancelled at this time. Similarly, the Climate Action Reserve adopts a 100-year commitment period for nature-based removal projects.

In the Engineered Carbon Removals sector, Puro.earth applies a 100-year time horizon for biochar projects, calculating issuance based on likely natural reversal when storing biochar on soil for 100 years. However, it remains unclear if adjustments would be made for reversals beyond this timeframe. Conversely, Carbfix In-situ Mineralisation and Drax X Stockholm Exergi methodologies claim Engineered Carbon Removals are 'permanent', suggesting negligible risk of reversal over any period. In these scenarios, the suggestion is that there is no risk of reversal and the carbon is indefinitely removed, with no commitment period identified.

Despite the assumed durability of Engineered Carbon Removal technologies, the lack of consistent commitment periods in the carbon market poses risks to project permanence. Assigning commitment periods allows for a direct comparison of storage capability between Engineered Carbon Removals and nature-based removal projects.

Our rating evaluates the removal and storage against the project's identified commitment period. In our view, the exclusion of commitment periods can create risk to project permanence for removal credits as it identifies a lack of storage commitment. Likewise, we find the use of elongated commitment periods that extend beyond human-relevant timeframes, e.g. 1000 years, creates permanence risk for carbon removal credits as a commitment to MRV over these timeframes is unlikely feasible. Even in cases where reversal risk is negligible, the lack of an assigned commitment period introduces risk.

Durability

Durability refers to the robustness of carbon storage. Durability is frequently used interchangeably with permanence to define this storage component. However, there is an important difference between the two. Permanence is a finite characteristic – CO₂ is either stored or it is not. Durability is a

characteristic subject to change. It recognises that the storage component is influenced by external factors, whether human or natural.

Durability is an important characteristic relevant only to removal credits, and not avoidance. In 2023, we published the Durability Assessment for carbon removal that evaluated the relative durability of six Engineered Carbon Removals and forestry from a top-down perspective. The assessment evaluated three components, the physical storage timeframe, the relative risk of reversal, and the underlying scientific certainty. The results are summarised in Figure 12.

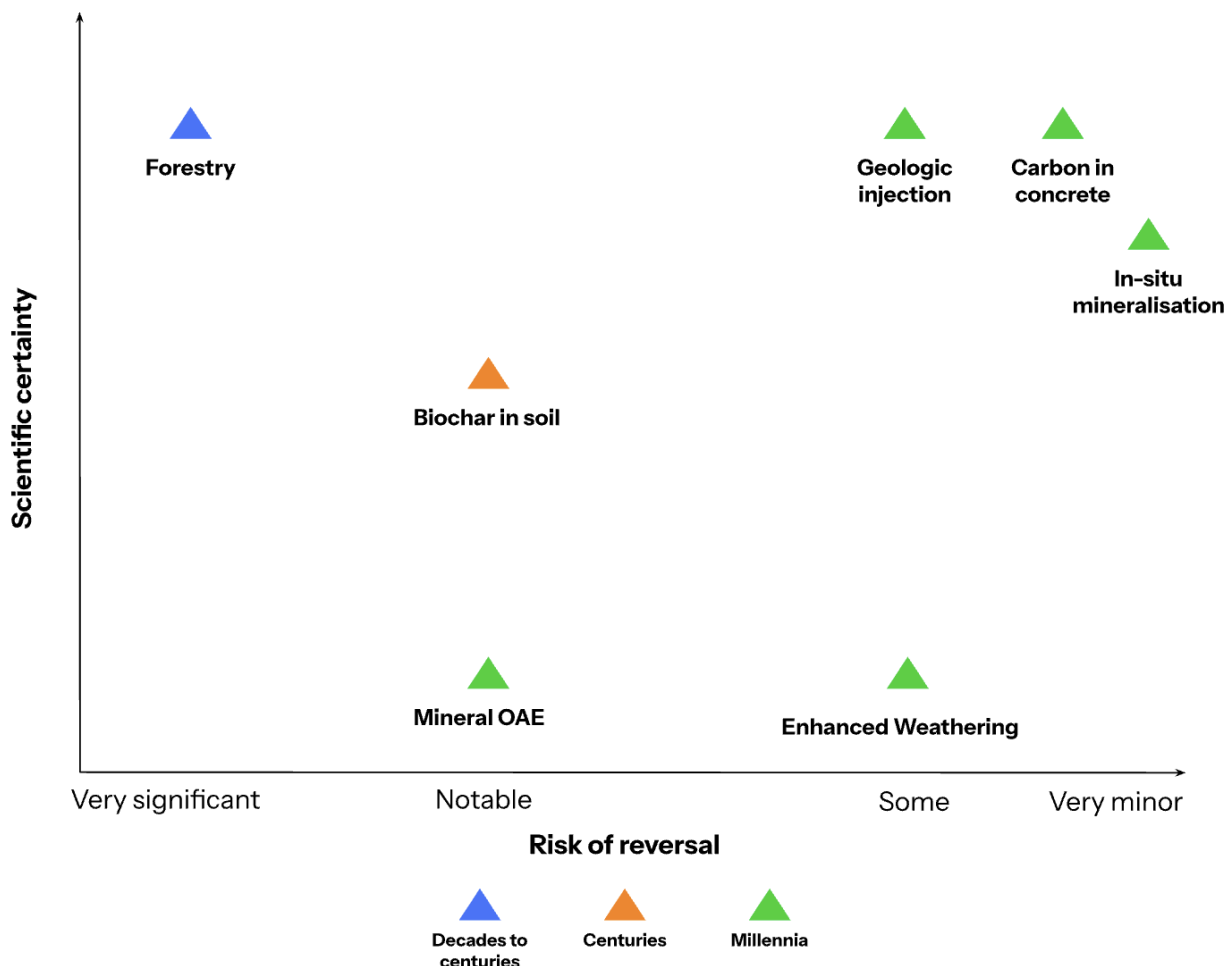


Figure 12. The durability of different carbon storage methods ranked by the risk of reversal and scientific certainty, from the 2023 BeZero Carbon [Durability Assessment](#)

The assessment highlights that in general Engineered Carbon Removals have greater durability than nature-based removals, with most methods evaluated having a greater period of physical storage and fewer risks of reversal. However, scientific certainty varies depending on the method.

For Engineered Carbon Removals, durability is an important characteristic in our assessment of permanence risk. However, durability is relative to the time frame against which it is being assessed. Over a nominal time period, such as a few decades, the durability of Engineered Carbon Removals would likely mitigate all permanence risks. Thus, our assessment of durability is informed by a project's commitment period.

Other risks

Natural risks

Natural risks are those introduced by activities outside of human control that could result in the reversal of carbon sequestration. For the Engineered Carbon Removals sector, these risks can extend from natural decay rates, changes in climate conditions, or extreme events. Some of these risks can be challenging to avoid, however, risk to carbon projects can likely be constrained through mitigatory activities.

Example 10. BeZero evaluates likely decay rates for biochar projects with soil application

To explore our assessment of natural risks, we can look at our analysis of currently rated biochar projects.

When applied to soil, biochar faces a natural reversal risk due to biological decay that occurs when exposed to natural climatic elements. Whereas when biochar is used as feedstock in concrete production, this reversal risk becomes negligible.

The durability of biochar has a large range in the literature, from decades to millennia. This spectrum is largely influenced by a project’s biological decay. The extent of decay is informed by project conditions, such as climate conditions in the area, the temperature of the soil the biochar is applied to, and importantly, the biological composition of the biochar. A characteristic of this composition is the molar hydrogen to organic carbon ratio (H/C_{org}) value. The lower the H/C_{org} value is, the greater the stability of the biochar and the lower the decay rate. The influence of this value has been analysed in scientific research and is presented in Figure 13.

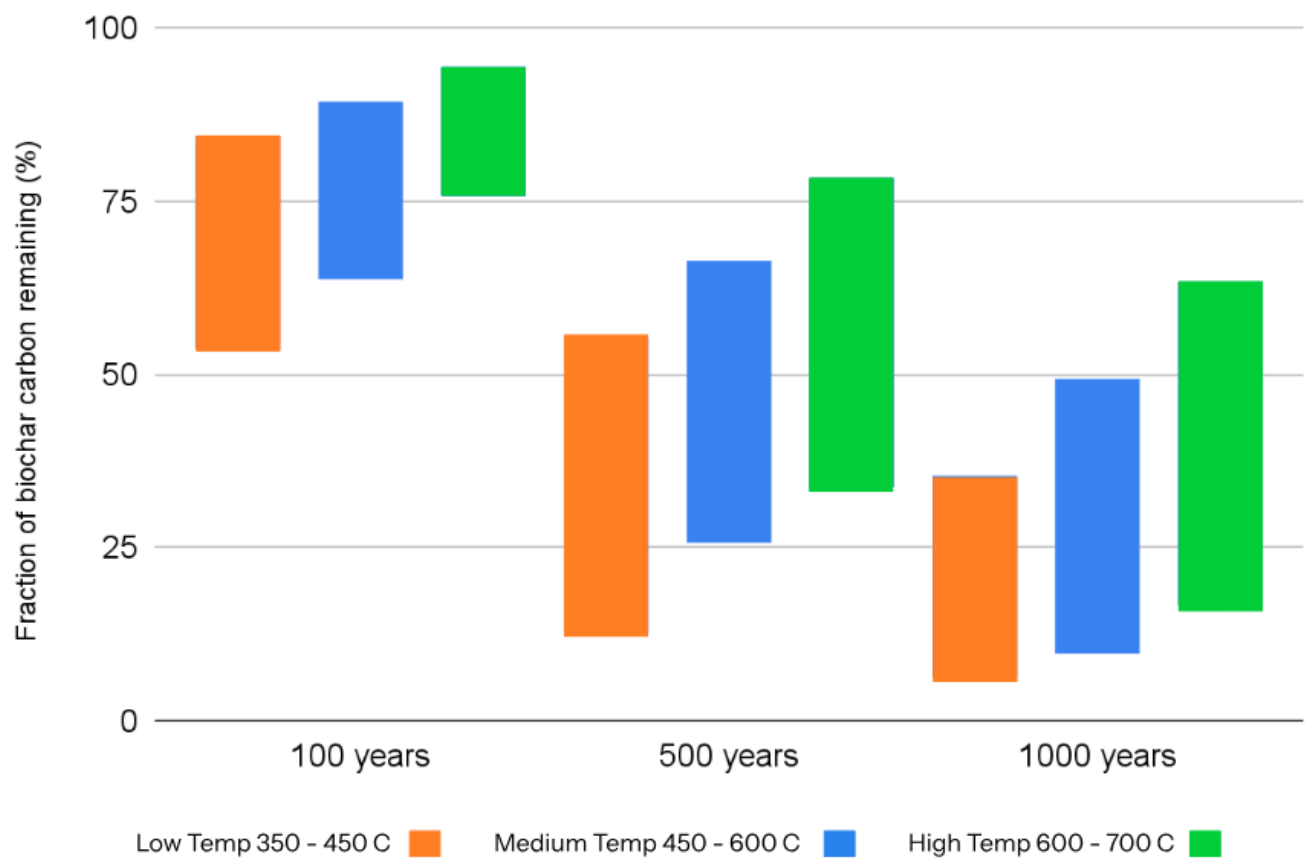


Figure 13. Biochar's decay rate is influenced by the pyrolysis conditions and the temperature of the soil to which it is applied. Higher pyrolysis temperatures of 600–700°C produce a more stable biochar. This research from Woolf et al. (2021) presents the decay rate for biochar with an H/C_{org} value in the range of 0.3–0.4. In general, the lower the H/C_{org} value, the more stable the biochar. The research also assessed decay rates for soil temperatures ranging from 5°C to 25°C, with colder temperatures reflected in the higher percentage of the fraction of biochar carbon remaining.

For currently rated biochar projects, the project conditions limited the impact of this natural risk. Both projects utilised pyrolysis temperatures between 600 and 700°C and used waste wood as the biomass feedstock, which produced H/C_{org} values of 0.16 and 0.17. Further, both applied their biochar to European soils of roughly 10°C. We found the conditions of both projects likely to produce a stable and durable biochar, with few risks of reversal due to the biological decay rate.

In general, the risk of biological decay can be tempered by the use of a realistic commitment period. If a project uses a 100-year commitment period, it is likely some decay could occur that can be accounted for in the crediting, introducing minor risks. However, if a 1000-year period is used, we would likely find significant risks due to biological decay.

Risk mitigation instruments

Stakeholder engagement

Ensuring communities' free, prior, and informed consent through a robust consultation process mitigates permanence risk as it imparts a level of community support. Voluntary community adherence to the project and involvement in project activities and achievements is fundamental for any project to succeed. Failing to secure this consent not only creates risk for the project's permanence but also leads to unfair outcomes for communities.

To conduct these evaluations, BeZero reviews project documentation and project-related external literature to ascertain whether a declaration of voluntary implementation exists.

Risk buffer and mitigation

Our assessment of permanence risk evaluates whether a project is mitigating the risks sufficiently through its contributions to a buffer pool and/or other mitigatory activities. For Engineered Carbon Removals, there are variations between sub-sectors as to which mitigatory activities are implemented.

Further, we find that some project developers choose to implement project-specific mitigatory activities, beyond methodology requirements, to employ a conservative approach where methodological or scientific uncertainties are present. For this reason, we assess mitigatory activities for each project as they are applied.

Impact of co-benefits and SDGs on carbon efficacy

Co-benefits and carbon efficacy

The BCR only assesses credit quality as it relates to carbon efficacy, and is not an assessment of any other element of the credit's quality, such as potential co-benefits from broader ecological and social impacts. These could include biodiversity effects; social, health, or economic impacts on local communities; or actual or potential Sustainable Development Goal (SDG) claims.

Data and reporting on co-benefits and SDG claims vary widely in the carbon market. In many cases, evidence on co-benefits is subject to less rigorous reporting, disclosure, and independent oversight than is the case for the carbon accreditation process.

Sustainable Development Goals

The UN SDGs are commonly used in the carbon market to illustrate the co-benefit impacts of carbon projects. Their use is becoming increasingly popular; over half of all projects in the carbon market made SDG claims in 2022.

As of January 2023, we have rated two projects in the Engineered Carbon Removals sector group. One of these projects claimed SDGs. Given this limited coverage, our rating process has uncovered few insights specific to this sector group. However, we note several challenges in assessing SDG claims across the carbon market and in the Engineered Carbon Removals sector.

- **Scalability of the indicators:** The SDG framework was designed for national reporting, not for use at the project level. Thus, many indicators are not fit for use at the project scale and must be adjusted by developers. For example, SDG 15 is commonly used to demonstrate biodiversity impacts in the carbon market but indicators such as 15.9.1(b) 'integration of biodiversity into national accounting and reporting systems, defined as the implementation of the System of Environmental Accounting', are not appropriate at the project scale.
- **Indicator impact representation:** Additionally, not all indicators represent the impacts that the goals are perceived to represent. For instance, three indicators for SDG 15 (15.9.1 (b), 15.a.1, and 15.b.1) are focused on economic impact rather than biodiversity impact. This misalignment with how the goal is perceived could lead to a misunderstanding of carbon market SDG claims.
- **Reporting inconsistencies:** Due to the indicator adjustments that often must be made, there are inconsistencies in how SDG impacts are measured. Additionally, there is a lack of standardisation amongst the systems standards bodies have for processing SDG claims. The systems range from relying on qualitative information provided by the project developer to requiring quantitative data, monitoring, and third-party validation and verification. However, all SDG claims appear the same on the surface. Projects are tagged with the same SDG symbol regardless of the SDG claim system the project underwent.
- **Inability to verify claims:** The data publicly available on project SDG impacts varies widely due to the lack of standardised requirements of standards bodies' SDG claim systems. The identification of impact metrics and the collection (through monitoring) of impact data from before the project started is essential for assessing the SDG impact a project has made. Nevertheless, projects often have little to no impact data available.

Interaction of carbon efficacy and co-benefits within Engineered Carbon Removals

For Engineered Carbon Removals projects, several specific co-benefits form part of our assessment of carbon efficacy, notably for additionality and leakage. These are summarised in Table 8.

Table 8. Co-benefits interact with our assessment of different risk factors in our assessment of Engineered Carbon Removals projects. Note that this list is not exhaustive.

Co-benefit	Risk factor	Analytical implication	SDG
Environmental safeguards	Additionality and carbon accounting	Improved industrial efficiency can reduce raw material usage and waste across the entire production process.	6, 11, 12, and 13
Reducing fossil fuel reliance	Additionality and leakage	Where projects produce renewable bioenergy reducing the need for fossil fuel-derived electricity.	7 and 12
Employment of local people	Additionality and carbon accounting	Where projects rely on carbon finance to deliver employment opportunities that are critical to delivering the project activities on the ground effectively.	1 and 8
Soil or forest health	Additionality, carbon accounting, and permanence	Where project activities remediate or improve terrestrial ecosystems.	15
Ocean health	Additionality, carbon accounting, and permanence	Where project activities remediate or improve marine ecosystems.	14

Appendix I: Analytical independence

BeZero Carbon acts as an independent third party and is not conflicted in delivering the BCR for the following reasons.

- BeZero Carbon's analysis and the resulting rating are limited to our assessment of the risks associated with the information provided in the public domain and expressed as a risk metric.
- BeZero Carbon does not provide any recommendations or advice on how to change or improve the project.
- BeZero Carbon does not create standards for, develop, invest, or transact in carbon projects. The only exception is the retirement of carbon credits for the explicit purpose of compensating for its own carbon footprint.
- BeZero Carbon does not verify, validate, sanction, or in any way influence the number of credits issued by the project.
- BeZero Carbon is not incentivised commercially or in any other sense to deliver a specific rating outcome at the time of the assignment or at any time in the future.
- All members of BeZero's analytical team, including the committee members, are commercially independent of the assigned ratings – i.e. their compensation, benefits, or performance measures are not in any manner linked to the ratings assigned.
- All BeZero staff, including all members of the ratings team, adhere to strict compliance procedures, including, inter alia, prohibition from holding and/or dealing in carbon credits and annual reporting. These standards are akin to standards practised by financial market rating agencies.
- BeZero Carbon has implemented a Rating Committee process, which mitigates the undue influence of individuals on the overall ratings process outcome.

Appendix II: Risk factor definitions and analytical framework matrix

Risk factor	aaa	aa	a	bbb	bb	b	c	d
Likelihood	Highest	Very high	High	Moderate	Moderately low	Low	Very low	Lowest
Additionality The likelihood that a credit purchased and retired leads to a tonne of CO ₂ e being avoided or sequestered that would not have otherwise happened.	The credit has the highest likelihood of being additional.	The credit has a very high likelihood of being additional.	The credit has a high likelihood of being additional.	The credit has a moderate likelihood of being additional.	The credit has a moderately low likelihood of being additional.	The credit has a low likelihood of being additional.	The credit has a very low likelihood of being additional.	The credit has the lowest likelihood of being additional.
Carbon Accounting The likelihood of having carbon accounting consistent with fully achieving a tonne of CO ₂ e avoided or removed.	The credit has the highest likelihood of having carbon accounting consistent with achieving a tonne of CO ₂ e avoided or removed.	The credit has a very high likelihood of having carbon accounting consistent with achieving a tonne of CO ₂ e avoided or removed.	The credit has a high likelihood of having carbon accounting consistent with achieving a tonne of CO ₂ e avoided or removed.	The credit has a moderate likelihood of having carbon accounting consistent with achieving a tonne of CO ₂ e avoided or removed.	The credit has a moderately low likelihood of having carbon accounting consistent with achieving a tonne of CO ₂ e avoided or removed.	The credit has a low likelihood of having carbon accounting consistent with achieving a tonne of CO ₂ e avoided or removed.	The credit has a very low likelihood of having carbon accounting consistent with achieving a tonne of CO ₂ e avoided or removed.	The credit has the lowest likelihood of having carbon accounting consistent with achieving a tonne of CO ₂ e avoided or removed.
Permanence The likelihood that the carbon avoided or removed by a project will remain so for the time committed.	The carbon avoided or removed by the credit has the highest likelihood of remaining permanent for the time committed.	The carbon avoided or removed by the credit has a very high likelihood of remaining permanent for the time committed.	The carbon avoided or removed by the credit has a high likelihood of remaining permanent for the time committed.	The carbon avoided or removed by the credit has a moderate likelihood of remaining permanent for the time committed.	The carbon avoided or removed by the credit has a low likelihood of remaining permanent for the time committed.	The carbon avoided or removed by the credit has a low likelihood of remaining permanent for the time committed.	The carbon avoided or removed by the credit has a very low likelihood of remaining permanent for the time committed.	The carbon avoided or removed by the credit has the lowest likelihood of remaining permanent for the time committed.

Appendix III: Sources as of January 2025

Third-party datasets

We also incorporate many third-party or industry datasets and sources into our assessments of individual risk factors. These include but are not limited to:

National and regional government information on industrial emissions, regulations, trade flows, production, consumption and commodity pricing; policy trackers for impact, legislation, governance and action; emission factor databases, such as from IGES; the NDC registry; carbon credit pricing databases; UN Least Developed Countries; World Bank governance indices; IPCC datasets for global warming potentials; Lawrence Livermore National Laboratory Roads to Removal; cdr.fyi dashboard; Ocean Visions database.

Literature sources

Our curated database of literature sources spans all types of peer-reviewed literature and industry research, with hundreds of entries for every sector, region, or risk factor.

The number of relevant literature-based sources specific to the Engineered Carbon Removals sector runs to over 420 as of January 2025. In Figure 14, we show how these sources are split by carbon risk factor.

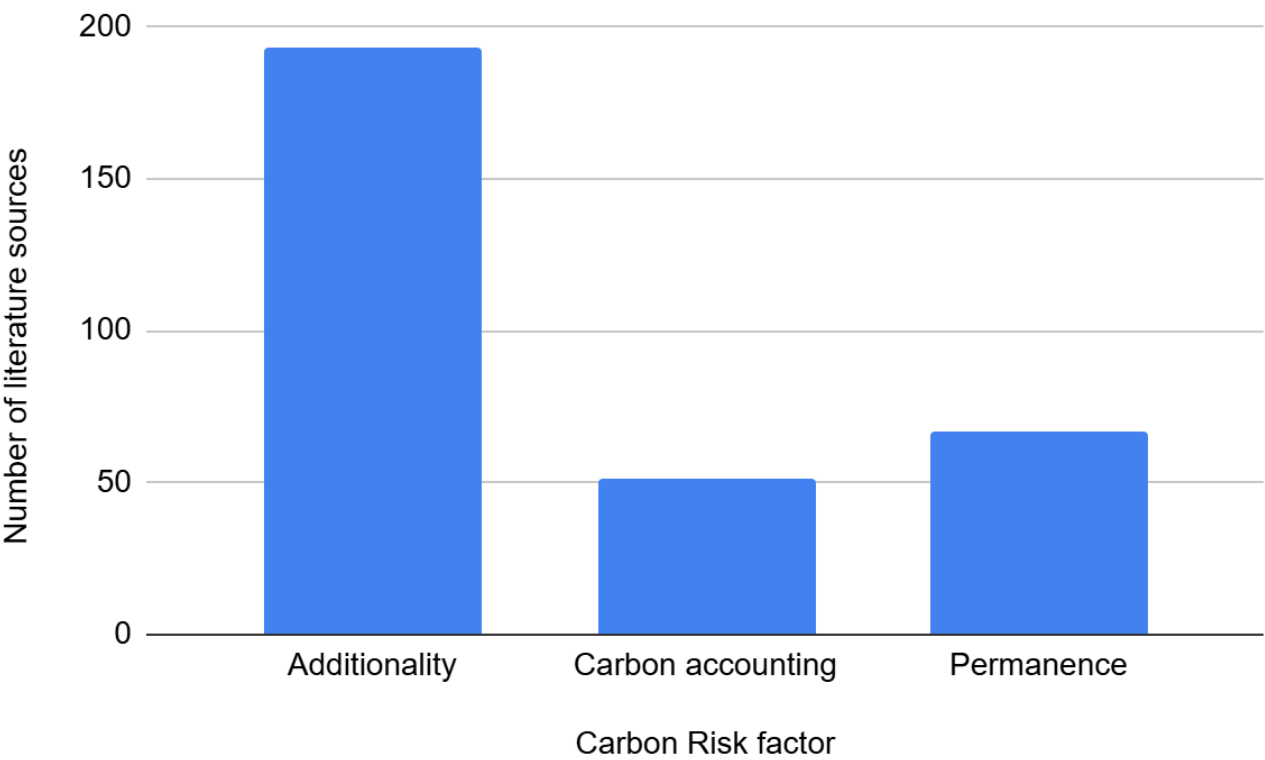


Figure 14. The number of literature sources relevant to the Engineered Carbon Removals sector used by BeZero Carbon Ratings as of January 2025.

Sample of literature sources

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Appendix IV: Additional reading

Check out the [ratings resources](#) page on our website to find links to all of our published methodologies, in addition to our series of risk factor assessment frameworks, our frameworks for assessing project methodologies and country-level risks, deep dives on factors influencing the carbon efficacy of projects in various sectors, and more.

Appendix V: Sustainable Development Goals

[Applying the equality SDGs to the carbon market](#)

[A focus on climate action: Sustainable Development Goal 13 claims in the carbon market](#)

[Transparency is key for SDG claims to be an effective asset in the carbon market](#)

[Interpreting SDG claims in voluntary carbon projects](#)

[Time to rethink biodiversity: SDG 14 & 15](#)

[Eye for detail: buyers want to know the evidence behind SDG claims](#)

[Lost in translation: SDG claims are more than meets the eye](#)

[How robust are SDG 3 & 7 claims in the carbon market?](#)

[Mapping the SDG claim lifecycle: 2023 update](#)

[Applying the equality SDGs to the carbon market](#)

[How economic SDG claims can be impactful in the carbon market](#)

Updates and reviews

Version number	Date	Description
1.00	26/02/24	Initial release
1.01	03/10/24	Clarification on additionality as a limiting factor
2.0	19/02/25	Updated risk factor framework
2.01	05/03/2025	Update to sector classification

Disclaimer

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